Resumen

Este Trabajo de Fin de Máster presenta un estudio sobre el diseño, fabricación y caracterización de celdas de vapor escritas por láser (LWVCs), sensores atómicos miniaturizados completamente de vidrio. Estos dispositivos confinan vapores de metales alcalinos en cavidades microfabricadas y son una plataforma prometedora para aplicaciones de fotónica integrada. La fabricación se basa en dos técnicas: el *Optical Contact Bonding*, proceso sin adhesivo que une superficies mediante fuerzas intermoleculares, y la microfabricación por láser de femtosegundo, que crea microestructuras 3D en el interior del vidrio. Ambas se complementan con tratamientos químicos, de plasma, térmicos y de presión. La caracterización incluye inspección visual mediante anillos de Newton, medición de energía de unión por el método de doble viga en voladizo, ensayos de tracción, análisis nanométrico por microscopía de fuerza atómica y pruebas de fugas por espectroscopía. Las aplicaciones exploradas incluyen magnetoencefalografía, relojes atómicos, instrumentación espacial e inspección de corrosión en vías de tren. Experimentos de magnetometría demuestran la detección de resonancia Hanle en una celda de vapor de cesio.

Palabras clave: Celdas de Vapor Escritas por Láser; Microfabricación por Láser de Femtosegundo; Unión por Contacto Óptico; Vapor de Metal Alcalino; MEMS; Fotónica Integrada; Activación de Superficies; Tratamiento de Plasma; Sensores Atómicos; Magnetometría; Resonancia Hanle; Magnetoencefalografía; Relojes Atómicos; Espectroscopía de Absorción; Hermeticidad al Helio; Microscopía de Fuerza Atómica; Proceso FLICE; Unión a Baja Temperatura

Resum

Aquest Treball de Fi de Màster presenta un estudi sobre el disseny, la fabricació i la caracterització de cel·les de vapor escrites per làser (LWVCs), sensors atòmics miniaturitzats completament de vidre. Aquests dispositius confinen vapors de metalls alcalins en cavitats microfabricades i constitueixen una plataforma prometedora per a aplicacions de fotònica integrada. La fabricació es basa en dues tècniques: l'*Optical Contact Bonding*, procés sense adhesiu que uneix superfícies mitjançant forces intermoleculars, i la microfabricació per làser de femtosegon, que crea microestructures 3D a l'interior del vidre. Ambdues es complementen amb tractaments químics, de plasma, tèrmics i de pressió. La caracterització inclou inspecció visual mitjançant anells de Newton, mesura d'energia d'unió pel mètode de doble biga en voladís, assajos de tracció, anàlisi nanomètrica per microscòpia de força atòmica i proves de fuites per espectroscòpia. Les aplicacions explorades inclouen magnetoencefalografia, rellotges atòmics, instrumentació espacial i inspecció de corrosió en vies de tren. Experiments de magnetometria demostren la detecció de la ressonància Hanle en una cel·la de vapor de cesi.

Paraules clau: Cel·les de Vapor Escrites per Làser; Microfabricació per Làser de Femtosegon; Unió per Contacte Òptic; Vapor de Metall Alcalí; MEMS; Fotònica Integrada; Activació de Superfícies; Tractament de Plasma; Sensors Atòmics; Magnetometria; Ressonància Hanle; Magnetoencefalografia; Rellotges Atòmics; Espectroscòpia d'Absorció; Hermeticitat enfront de l'Heli; Microscòpia de Força Atòmica; Procés FLICE; Unió a Baixa Temperatura

Abstract

This Master's thesis presents a study on the design, fabrication, and characterization of Laser-Written Vapor Cells (LWVCs), a novel class of miniaturized all-glass atomic sensors. These devices confine alkali metal vapors in micro-machined cavities and are promising platforms for integrated photonics applications. Fabrication relies on two core techniques: Optical Contact Bonding, a glueless process joining glass surfaces through intermolecular forces, and Femtosecond Laser Micromachining, which creates buried 3D microstructures inside glass using ultrashort laser pulses. Both are complemented by chemical, plasma, thermal, and pressure treatments. Characterization methods include visual inspection via Newton's rings, bonding energy measurements using the double cantilever beam method, tensile strength testing, sub-nanometric surface analysis via Atomic Force Microscopy, and a spectroscopy-based leak test protocol. Applications explored include magnetoencephalography, atomic clocks, space instrumentation, and rail corrosion inspection. Preliminary magnetometry experiments demonstrate Hanle resonance detection in a cesium vapor cell, validating the potential of LWVCs as compact, high-performance atomic sensors.

Keywords: Laser-Written Vapor Cells; Femtosecond Laser Micromachining; Optical Contact Bonding; Alkali Metal Vapor; MEMS; Integrated Photonics; Surface Activation; Plasma Treatment; Atomic Sensors; Magnetometry; Hanle Resonance; Magnetoencephalography; Atomic Clocks; Absorption Spectroscopy; Helium Hermeticity; Atomic Force Microscopy; FLICE Process; Low-Temperature Bonding

RESUMEN EJECUTIVO

La memoria del TFM del MUQIP debe desarrollar en el texto los siguientes conceptos, debidamente justificados y discutidos, centrados en el ámbito de la ELEGIR DISCIPLINA

CONCEPT (ABET)	**CONCEPTO (traducción)**	**¿Cumple? (S/N)**	**¿Dónde? (páginas)**
1. IDENTIFY:	1. IDENTIFICAR:		
1.1. Problem statement and opportunity	1.1. Planteamiento del problema y oportunidad		
1.2. Constraints (standards, codes, needs, requirements & specifications)	1.2. Toma en consideración de los condicionantes (normas técnicas y regulación, necesidades, requisitos y especificaciones)		
1.3. Setting of goals	1.3. Establecimiento de objetivos		
2. FORMULATE:	2. FORMULAR:		
2.1. Creative solution generation (analysis)	2.1. Generación de soluciones creativas (análisis)		
2.2. Evaluation of multiple solutions and decision-making (synthesis)	2.2. Evaluación de múltiples soluciones y toma de decisiones (síntesis)		
3. SOLVE:	3. RESOLVER:		
3.1. Fulfilment of goals	3.1. Evaluación del cumplimiento de objetivos		
3.2. Overall impact and significance (contributions and practical recommendations)	3.2. Evaluación del impacto global y alcance (contribuciones y recomendaciones prácticas)		

”¡Bien, riámonos porque Don Quijote toma a los molinos por gigantes! ¡Bien, Don Quijote no debió ver un gigante donde había un molino! Pero, ¿por qué el hombre sabe de gigantes? ¿Dónde hay o ha habido gigantes? Mas, si no los hay ni los ha habido, resulta que no Don Quijote, sino el hombre, la especie humana en algún momento de su historia descubrió un gigante donde no lo había”.

– **Antonio Machado**, *Juan de Mairena*.

Contents

List of Figures

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List of Abbreviations

AFM Atomic Force Microscopy.

FEM Finite-element method.

FLICE Femtosecond Laser Induced Chemical Etching.

FLW Femtosecond Laser Writing.

FWHM Full Width Half Maximum.

HWP Half Wave Plate.

LWVC Laser-Written Vapor Cell.

MEMS Micro-electromechanical systems.

NEG Non-evaporable Getter.

PBS Polarizing Beam Splitter.

QWP Quarter Wave Plate.

RIE Reactive Ion Etching.

RMS Root Mean Square.

TTV Total Thickness Variation.

Part I

Thesis

Chapter 1

Introduction and Motivation

1.1 Historical Evolution

A sensor is an analytical device designed to detect and respond to a specific input stimulus from the physical environment, translating it into a measurable analog or digital signal. These input stimuli encompass a vast array of environmental phenomena, such as pressure, force, flow, light, heat, motion, or moisture. The resulting output is typically converted into an electrical signal that can be subsequently processed, measured, and analyzed [**javaid2021**].

In order to better understand physical reality, a wide variety of sensors have been developed throughout human history, with some of the earliest documented mechanisms dating back nearly two millennia. Early sensing principles relied entirely on mechanical, hydraulic, and pneumatic feedback loops rather than electronic transduction. For instance, Heron of Alexandria developed an automatic mechanism for opening and closing temple doors in 62 CE. This device utilized the thermal expansion of air to trigger mechanical motion, as illustrated in Figure 1.1. Similarly, Zhang Heng's seismoscope (132 CE) employed an inertia-based internal pendulum to detect and indicate the direction of distant earthquakes, as detailed in Figure 1.2.

The precursor to modern quantitative sensors began to emerge during the 17th century. In contemporary engineering, these devices commonly rely on three key interconnected elements: (i) a receptor that specifically interacts with the target analyte or physical stimulus; (ii) a transducer that responds to this interaction by producing a characteristic signal; and (iii) a data-processing system that computes the signal into measurable quantities and generates usable data [5]. As one of the earliest examples of this evolving conceptual framework, Galileo Galilei invented the thermoscope, which monitored temperature variations through the thermal expansion of air. This qualitative device was subsequently improved by Santorio Santorio, who introduced a numerical scale, effectively transforming the instrument into the first measurable thermometer [6].

During the Industrial Revolution (18th to 19th century), the scientific imperative to analyze and optimize machine behavior demanded more diverse and sophisticated monitoring tools. It was during this era that the foundational concept of transduction—translating a physical phenomenon into an actionable, measurable output—was firmly established. Concurrently, essential analytical tools such as the barometer, litmus paper, the Davy safety lamp, and early mechanical thermostats were developed to ensure industrial safety and process control [5, 7].

The 19th century marked a critical turning point in sensor technology by shifting the output medium from purely mechanical displacements toward electrical signals. The invention of the galvanometer

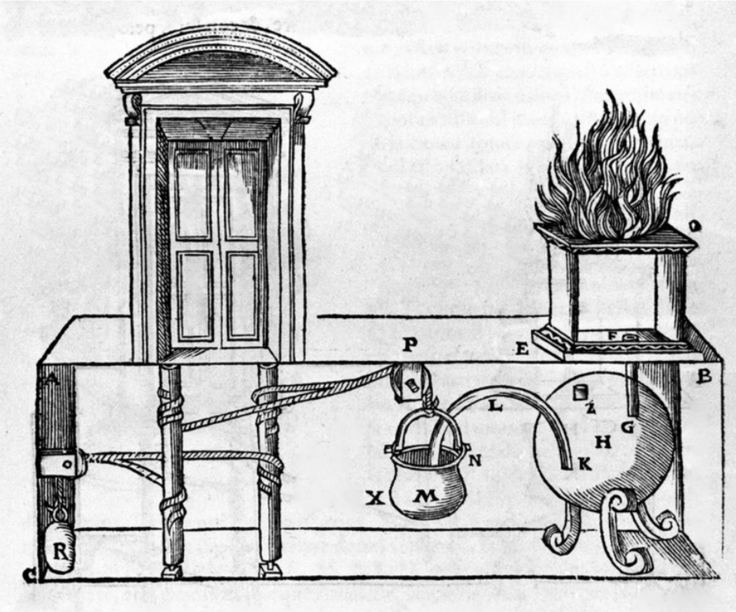

Figure 1.1: Automatic mechanism for temple door actuation designed by Heron of Alexandria (62 CE). When the altar fire is lit, the air confined within cavity (E) undergoes thermal expansion, increasing the pneumatic pressure inside the water-filled reservoir (H). This pressure forces water through pipe (L) into the suspended cauldron (M). The resulting weight displacement overcomes the counterweight (R), thereby opening the doors. Once the fire is extinguished, pressure drops, and the system reverts to its initial equilibrium state. Adapted from [3].

by Johann Schweigger in 1820 allowed for the precise measurement of micro-currents, effectively paving the way for electrical instrumentation. Later, in 1880, Pierre and Jacques Curie discovered the piezoelectric effect, a breakthrough that enabled the development of piezoelectric sensors capable of converting mechanical stress directly into measurable electrical charge [8].

The mid-20th century witnessed the introduction of solid-state electronic sensors, transforming global industries by providing faster, more reliable, and highly accurate measurements. This era was revolutionized by the invention of the point-contact transistor at Bell Labs in 1947, which enabled the development of significantly smaller and more efficient electronic components [9]. Following World War II, the rapid growth of the semiconductor industry allowed sensors to become progressively more compact, efficient, and affordable. This technological trajectory was further propelled by the introduction of integrated circuits (ICs) in the late 1950s and 1960s, a breakthrough that facilitated the micro-miniaturization of devices and enabled the integration of multiple sensing functions onto a single silicon substrate [10]. By the 1960s and 1970s, these advanced solid-state sensors became widespread across the aerospace, automotive, and telecommunications sectors, ultimately forming the technological backbone of complex, modern monitoring and control systems. This relentless drive toward miniaturization and integration set the stage for an entirely new class of devices capable of combining mechanical and electronic functionality at the microscale.

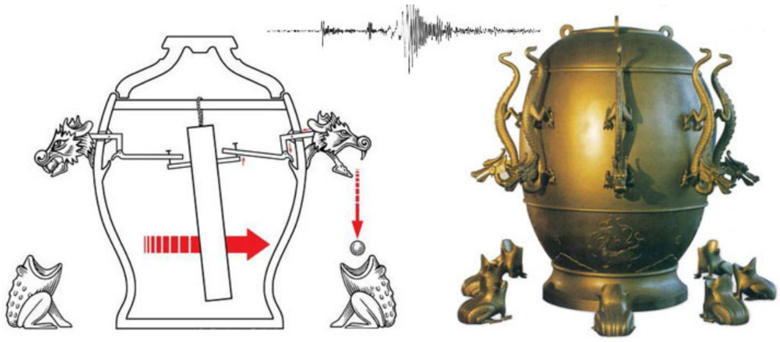

Figure 1.2: Seismoscope developed by Chinese polymath Zhang Heng (132 CE). A central pendulum suspended inside the bronze vessel oscillates in response to seismic vibrations. This displacement triggers an internal lever mechanism that opens the jaws of a specific dragon head, releasing a bronze ball into the mouth of a corresponding frog statue below. Eight radial dragon-frog pairs were meticulously aligned to determine the primary orientation of the earthquake epicenter. Adapted from [4].

1.2 Microelectromechanical Systems (MEMS)

Microelectromechanical Systems (MEMS) refer to devices with characteristic lengths between 1 µm and 1 mm (see Figure 1.4) that combine electrical and mechanical components fabricated using integrated circuit batch-processing technologies [11]. In recent decades, MEMS have found widespread application across a variety of fields, driven by several inherent advantages of the technology.

Silicon, the primary structural and active material, is Earth-abundant, cost-effective, and can be produced to extraordinary levels of purity and crystalline perfection. Its processing relies on thin deposited films that are intrinsically amenable to miniaturization, while photolithographic patterning techniques enable highly precise and reproducible definitions of device geometries at microscale dimensions. Perhaps most significantly from a commercial standpoint, MEMS devices benefit from the same batch-fabrication paradigm as microelectronic circuits: a single semiconductor wafer simultaneously yields hundreds of identical devices, as exemplified in Figure 1.3, dramatically reducing per-unit costs and enabling large-scale deployment [**petersen1982**].

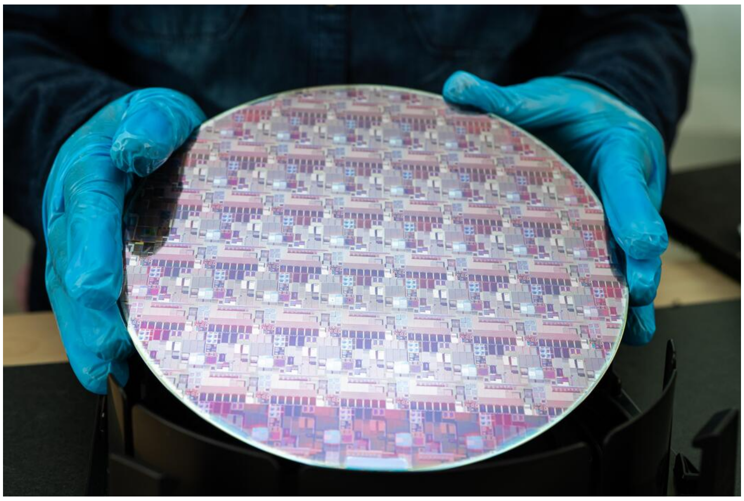

Figure 1.3: Example of a batch-fabricated semiconductor wafer. Parallel manufacturing yields hundreds of identical micro-devices simultaneously, significantly lowering per-unit production costs.

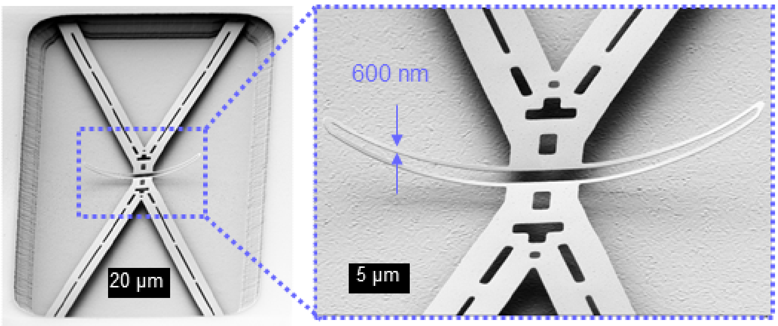

Figure 1.4: Structural representation of a MEMS device. Microstructures typically exhibit characteristic dimensions on the order of micrometers [12].

The exceptional purity and crystalline perfection achievable in silicon substrates translate directly into optimized mechanical properties, much as these same characteristics have long been exploited to maximize electronic performance, reliability, and reproducibility in integrated circuits [**petersen1982**]. As a consequence, MEMS devices have attained levels of precision, sensitivity, and long-term stability that were previously unachievable in macroscopic mechanical systems, enabling their widespread adoption in consumer electronics, biomedical instrumentation, environmental monitoring, and precision metrology.

1.3 Atomic Sensors

The demand for precise and stable measurements has driven the development of quantum sensors based on atomic spectroscopy. Atom-based quantum sensors exploit the highly stable, reproducible, and sensitive energy levels of atoms to measure physical quantities [2, 1]. In their basic form, these sensors create a specific quantum state in a gas-phase atom using laser light or electromagnetic fields. They then measure a property of that state to determine the value of the physical quantity being tested.

At their core, all atomic sensors share three main components: a light source, a collection of identical atoms, and an optical method for detection, as shown in Figure 1.5. Modern designs use high-performance diode lasers, electro-optic modulators, and sensitive photodetectors to reach fundamental quantum limits.

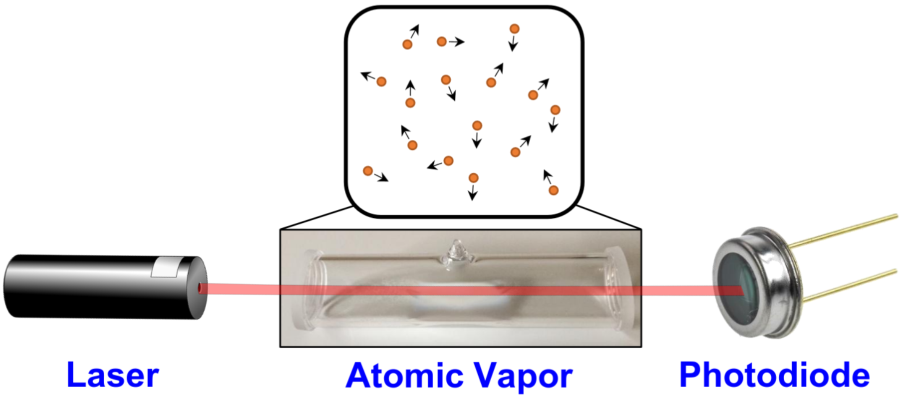

Figure 1.5: Essential elements of an atom-based sensor: a laser light source, an atomic vapor cell, and a photodiode detector. A resonant beam passes through the cell; the transmitted (or fluorescent) light carries information about the atomic quantum state and the physical quantity of interest. Adapted from [1].

1.3.1 Vapor Cell Implementation

The most common and compact setup for an atomic sensor is the vapor cell. This is a sealed glass container filled with an alkali-metal gas [2]. A laser beam tuned to one of the optical transitions of the atoms is shone through the cell, and a photodetector monitors the transmitted light or fluorescence. This simple setup allows researchers to control and read the quantum properties of atoms with high accuracy while keeping the device small.

Inside the cell, the alkali atoms move fast due to thermal energy and constantly collide with the walls. To keep the quantum spin state alive longer (extending the coherence time T_2 beyond the microsecond limit caused by these collisions), two main methods are used [1].

The first method fills the cell with an inert buffer gas mixture, such as nitrogen, helium, or neon. This gas slows down the movement of the atoms toward the walls without destroying their spin state. This approach keeps the quantum state stable for tens of milliseconds in centimeter-sized cells.

The second method coats the inner walls of the cell with special materials like paraffin or alkene-based substances. These coatings allow the atoms to bounce off the walls thousands of times without losing their spin state, pushing T_2 close to one second.

The choice between these methods depends on the application, with vapor cells offering the best balance of size, weight, and power (SWaP) for magnetometers and basic atomic clocks.

1.3.2 Measurement Principle

All atomic sensors are fundamentally based on measuring the energy difference between discrete atomic states [2]. When an atom interacts with an external field, its energy levels shift. This shift can be precisely measured to find the exact strength of the field. The measurement cycle follows three main steps:

1. **State preparation.** The atoms are put into a well-defined quantum state using optical pumping. In this process, circularly polarized laser light transfers angular momentum to the atoms, lining up their spins along the laser beam axis. This establishes the initial baseline vector on the Bloch sphere before the system is exposed to external perturbations.

2. **State evolution.** The prepared state changes over time under the total energy of the system, described by the Hamiltonian $\hat{H} = \hat{H}_0 + \hat{H}_{\text{int}}$. Here, $\hat{H}_{\text{int}}$ represents the interaction with the external field being measured [1]. The quantum state builds up a phase shift:

$$\phi_{\text{int}} = \frac{1}{\hbar} \int H_{\text{int}} \, \mathrm{d}t = \frac{\mu F}{\hbar} T, \tag{1.1}$$

   where μ is the coupling parameter, F is the field strength, and T is the interaction time.

   As cited in the literature, for an atomic magnetometer, F corresponds to the external magnetic field B and μ is the magnetic moment of the atom. Conversely, in the case of inertial sensors, F represents the physical rotation rate Ω and μ is the angular momentum of the atom.

3. **Readout.** The final state is checked optically by measuring the transmission, fluorescence, or polarization change of a probe laser beam, which reveals ϕ_{int} and F.

1.3.3 Advantages of Atomic Sensors

Atoms are ideal for sensing because they meet four key requirements: they are identical, isolatable, interfaceable, and understandable [1]. First, all atoms of a specific isotope are perfectly identical. This means a well-designed sensor does not suffer from manufacturing variations in its sensing element.

Second, atomic energy levels and their reactions to external fields are very well known, allowing engineers to control and minimize systematic errors. For example, the core clock transition of cesium is independent of magnetic fields to first order, and warm gases in glass cells are largely unaffected by energy shifts from wall collisions.

Third, atoms are easy to interface with, as their strong interaction with resonant photons allows users to control and read their states using compact diode lasers and detectors.

Finally, atomic interactions with light and fields are understandable because they can be calculated from first principles with high accuracy, allowing realistic simulations without relying on guesswork.

1.3.4 Choice of Atomic Species

The vast majority of practical atomic sensors use heavy alkali atoms, specifically rubidium (Rb) and cesium (Cs) [1]. Their simple, hydrogen-like electron structure is well documented, providing exact data on transition frequencies and collision behaviors [13]. Narrow-linewidth diode lasers are readily available for the primary optical transitions of both elements, which greatly simplifies the optical setup.

Both elements create a stable vapor at low pressure and moderate temperatures between 80 and 130 degrees Celsius, allowing the use of sealed glass cells without complex vacuum pumps. Additionally, their ground-state hyperfine splittings can be directly measured using standard, high-performance microwave electronics.

For the most demanding applications, like advanced optical clocks and high-precision inertial sensors, alkaline-earth atoms like strontium or calcium are preferred. Their narrow optical transitions offer extremely high quality factors, though they require more complex laser systems.

1.3.5 Quantum Noise and the Standard Quantum Limit

The fundamental noise floor of any atomic sensor comes from quantum projection noise [2, 1]. Each measurement forces an atom into a specific energy state based on quantum probabilities, introducing unavoidable randomness between measurements. For N independent atoms with a coherence time T_2, the noise floor follows the Standard Quantum Limit:

$$\delta S \approx \frac{\partial S}{\partial E} \frac{\hbar}{\sqrt{N T_2}} \qquad \left[\frac{\text{units of } S}{\sqrt{\text{Hz}}}\right], \tag{1.2}$$

where $\partial S/\partial E$ shows how much the measured signal S changes with an energy shift. This equation highlights the main design trade-off: using more atoms (N) or achieving a longer coherence time (T_2) lowers the noise floor.

For short measurement times ($T \ll T_2$), the smallest detectable field depends on the phase resolution of the readout system and the coupling strength μ between the field and the atom:

$$\delta F_{\text{min}} = \frac{\hbar\, \delta\varphi}{\mu T}. \tag{1.3}$$

When the measurement runs longer than the coherence time ($T > T_2$), the sensitivity is calculated by adding up multiple independent measurements of length T_2. This yields a minimum detectable field that improves with the total averaging time τ:

$$\delta F_{\text{min}} = \frac{\hbar}{\mu \sqrt{N T_2 \tau}}. \tag{1.4}$$

In practice, technical noise sources—like laser intensity shifts, frequency drift, or imperfect magnetic shielding—usually dominate over quantum noise in real-world devices. Minimizing these everyday noise sources remains the primary engineering task.

1.3.6 Types of Atomic Sensors and Achievable Sensitivities

Table 1.1 summarizes the main types of atomic sensors, how they work, and their expected quantum-noise-limited sensitivities for centimeter-sized devices [2, 1].

Table 1.1: Key atomic sensor categories, measured quantities, and quantum-projection-noise-limited sensitivities for cm-scale setups. LPAI: light-pulse atom interferometer; OPM: optically pumped magnetometer. Adapted from [2, 1].

Sensor type	Quantity	Core Mechanism	QPN sensitivity
Atomic Clocks	Frequency	Hyperfine/Electronic transitions	$\sim 10^{-13}$ to $10^{-17}\,\mathrm{Hz}^{-1/2}$
OPM (SERF)	Magnetic field B	Zeeman Larmor precession	$\sim 10\,\mathrm{aT}/\sqrt{\mathrm{Hz}}$
Inertial (LPAI)	Accel. a / Rot. Ω	Matter-wave interference	$\sim 4\,\mathrm{ng}$ / $3 \times 10^{-4}\,{}^\circ/\sqrt{\mathrm{h}}$
Rydberg Sensor	Electric field E	Atomic Stark shift ($n \sim 50$)	$\sim 15\,\mathrm{pV\,cm}^{-1}/\sqrt{\mathrm{Hz}}$

Among these technologies, optically pumped magnetometers (OPMs) are the most mature vapor-cell sensors [2]. These devices track how atomic spins rotate in an external magnetic field after being lined up. Circularly polarized laser light aligns the atomic spins, and their natural rotation (Larmor precession) is driven and measured using optical techniques.

OPMs operating in the spin-exchange relaxation-free (SERF) regime stop the line-broadening usually caused by collisions between atoms, allowing them to reach incredible sensitivities below $1\,\mathrm{fT}/\sqrt{\mathrm{Hz}}$. This matches the performance of superconducting SQUID magnetometers but completely avoids the need for expensive cryogenic cooling.

Electric fields can also be measured with very high sensitivity in atomic gases by using Rydberg atoms. Rydberg states occur when an electron is excited to a very high energy level. This gives the atom a huge electric dipole moment, making it highly sensitive to energy shifts via the Stark effect when an electric field is present.

Using multiple lasers to create quantum paths between the ground state and these high Rydberg states allows researchers to read out field variations as visible changes in light transmission through room-temperature gases [1]. By measuring these shifts with microwave or laser tools, electric field sensitivities of $10\,\mu\mathrm{V}/\mathrm{cm}$ or better are standard.

In contrast to small vapor cells, light-pulse atom interferometers offer the highest performance for measuring acceleration and rotation. However, they require laser-cooled atoms and complex high-vacuum chambers, which significantly increases their size, weight, and power demands.

Today, field work in magnetometry and electrometry relies heavily on compact vapor cells, while the highest-performing gravimeters, gyroscopes, and clocks use ultra-cold atoms. Laser-written vapor cells, the focus of this thesis, offer a clear path to building miniature, mass-producible sensors that keep the core performance advantages of quantum technology.

1.4 Laser-Written Vapor Cells

explicar lwvcs + conectar en PICs

Chapter 2

Design of Vapor Cells

Chapter 3

Fabrication Methods

Precision miniaturized sensors have become an increasingly relevant technology over recent decades, finding applications in navigation, medical diagnostics, geophysical surveying, and fundamental physics research [14]. The growing demand for compact, high-performance devices has driven the development of new fabrication techniques capable of producing smaller, more complex geometries while maintaining extreme gas purity, free of contaminants that could degrade sensor performance. Manufacturing such devices is inherently challenging: the internal cavities must be hermetically sealed to prevent outgassing or contamination, the interaction volume must be precisely defined to ensure reproducibility, and the cell must accommodate optical access from multiple directions to support the beam geometries required by techniques such as multipass cells, dual-beam, or cavity-enhanced sensing [14]. Traditional fabrication strategies, such as MEMS-based anodic bonding, offer excellent scalability but are fundamentally limited to planar geometries and restricted optical access. Photolithographic approaches, while capable of producing nanostructured vapor cells, share the same limitation of planar bonding strategies [15].

A Laser-Written Vapor Cell (LWVC) addresses these limitations by exploiting the three-dimensional structuring capabilities of Femtosecond Laser Writing (FLW), enabling arbitrary buried geometries within a monolithic transparent substrate [14]. The complete fabrication workflow is illustrated in Fig. 3.1 and consists of four main steps, each of which contributes to the precision and hermeticity of the final device, as described below and discussed in detail in the following sections.

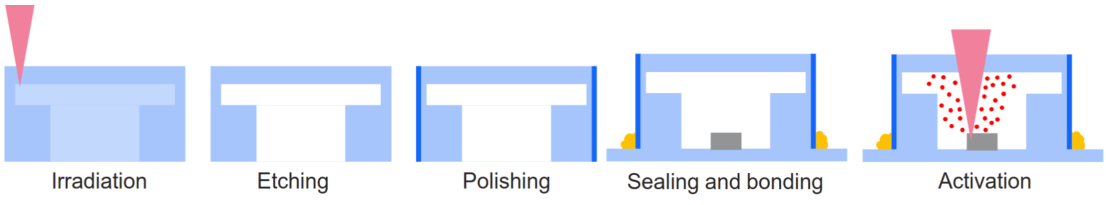

Figure 3.1: Fabrication worklow of a Laser-Written Vapor Cell. Steps (1-2) describe the FLICE process. In step (4) NEG dispenser placement and bonding are performed. Activation is illustrated by step (5). From [14].

Step 1: Femtosecond Laser Writing and Chemical Etching (FLICE)

The first step employs the so-called FLICE technique (Femtosecond Laser Irradiation followed by Chemical Etching), a two-stage maskless microfabrication process [14]. An ultrafast laser beam is tightly focused inside the bulk of a fused silica substrate, inducing the local formation of self-oriented nanogratings through nonlinear absorption. Because the material modification is confined to the laser focal spot, arbitrary three-dimensional patterns can be inscribed with micrometric resolution and without any lithographic mask. In the second stage, the substrate is immersed in a hydrofluoric acid solution, which selectively removes the irradiated material, leaving behind hollow channels and reservoirs with the desired geometry. The non-irradiated fused silica remains unaffected, preserving the monolithic integrity of the substrate and its broad optical transparency window—a key advantage for spectroscopic characterization of the gas content. This maskless approach enables geometries that are inaccessible to planar bonding or photolithographic techniques, including fully buried sensing channels and multi-sided optical access. The FLICE process and its optimization for LWVC fabrication are further discussed in Section 3.1.

Step 2: Placement of the Non-Evaporable Getter Dispenser

Once the internal cavities have been formed, a pill of non-evaporable getter (NEG) material is placed inside the reservoir cavity of the fused silica structure [14]. NEG materials serve a dual function: upon thermal activation, they simultaneously release the alkali metal vapor required for atomic sensing, and absorb residual buffer gases such as nitrogen. This dual action enables the production of cells with sub-atmospheric buffer gas pressures without the need for a vacuum apparatus, greatly simplifying the manufacturing process while maintaining the gas purity essential for high-precision applications. The ability to control the buffer gas pressure post-sealing—by adjusting the duration of getter heating or by pre-filling with controlled fractions of ungetterable gases such as argon—offers an additional degree of freedom for optimizing sensor performance [14].

Step 3: Sealing and Bonding

After the dispenser has been placed, the cell must be hermetically sealed to prevent contamination and ensure long-term stability. The sealing is performed inside a controlled-atmosphere chamber filled with nitrogen containing very low amounts of O_2, thereby preventing the trapping of reactive oxygen inside the cell [14]. The choice of bonding method is critical, as it directly affects hermeticity, thermal stability, and compatibility with the cell's optical coatings and internal components.

Bonding glass and crystal presents unique engineering challenges due to the material's non-porous nature, extreme smoothness, and high surface energy. In industrial applications, selecting the right adhesive or process is critical for ensuring optical clarity, structural integrity, and durability against environmental factors. Some of the most common glass bonding methods are:

- **UV-Curing Adhesives:** These are one-component resins that remain liquid until exposed to specific wavelengths of ultraviolet light. Upon UV exposure, a rapid polymerization process begins. This method offers strong, precise, and optically transparent joints, since the refractive index of the adhesive can be tuned to match that of the substrate [16].
- **Anodic Bonding:** The substrates are heated to 300°C–500°C while a high voltage (≈ 1 kV)

is applied. Ions in the glass migrate, creating a space-charge region that generates a strong electrostatic attraction, resulting in a permanent chemical bond [17].

- **Laser Microwelding:** A focused laser locally melts and fuses the glass substrates at the interface, creating a bond without the need for intermediate adhesive layers or high global temperatures [18].

While these methods are widely used, they present specific challenges in the context of MEMS and vapor cell manufacturing. Adhesives at the bonding interface can migrate into the sealed cavity and contaminate the gas, critically affecting the performance of high-precision devices such as atomic clocks [19]. Furthermore, adhesives may overheat upon interaction with high-power lasers during subsequent processing or operation steps, inducing mechanical stress, birefringence, thermal expansion, and crack formation [20]. Adhesive-based bonds also typically fail to meet the hermeticity standards required for demanding applications [21]. Adhesive-free methods such as anodic bonding, while achieving excellent hermeticity, require elevated global temperatures [17], which increases the risk of thermally induced stress and renders them incompatible with temperature-sensitive optical coatings [22].

A modern solution that can address all of these challenges simultaneously is *Optical Contact Bonding* [23], which achieves a hermetic, adhesive-free joint at room temperature through van der Waals and hydrogen-bond interactions between two ultra-flat, clean surfaces. This method is fully compatible with the FLICE-based LWVC fabrication process and is discussed in detail in Section 3.2.

Step 4: Getter Activation

In the final step, the sealed cell is exposed to a focused laser beam directed at the NEG dispenser through the fused silica substrate. The laser locally heats the dispenser, triggering the release of alkali vapor into the sensing chamber and the simultaneous absorption of residual buffer gas [14]. Because the activation is performed optically and post-sealing, the cell does not need to be re-opened or placed in a vacuum system, preserving hermeticity throughout the entire process. The precise control of activation time and laser power allows the residual buffer gas pressure to be tuned, directly influencing key sensing parameters such as the spin coherence time and pressure-induced spectral broadening. The activation procedure and its effects on cell performance are discussed in Section 3.3.2.

3.1 Femtosecond Laser Micromachining

The fabrication of vapor cells and other precision optical devices requires the controlled removal of glass to create internal cavities, channels, and complex three-dimensional structures. This task is non-trivial: glass is simultaneously transparent, brittle, and highly chemically resistant, which means that conventional machining approaches often fail to meet the precision and surface quality standards demanded by atomic sensing applications [24].

Mechanical machining with diamond-tipped tools or ultrasonic drilling is effective at the macro-scale but introduces micro-cracks and residual internal stress that can lead to structural failure in thin substrates [24]. Wet chemical etching with hydrofluoric acid (HF) is isotropic, meaning it attacks the material equally in all directions, making it difficult to produce deep channels with high aspect ratios or well-defined geometries. Dry plasma-based techniques such as Reactive Ion Etching (RIE) offer greater directionality but are restricted to surface patterns due to their extremely slow volumetric etch rates [25]. Critically, none of these methods allows for the fabrication of fully *buried* three-dimensional structures within a monolithic substrate, which is precisely the capability required by Laser-Written Vapor Cells.

3.1.1 Femtosecond Laser Writing

Femtosecond Laser Writing (FLW) overcomes all of the above limitations by exploiting the nonlinear optical response of transparent materials to ultrashort laser pulses. Glass is nominally transparent to visible and near-infrared light; however, when a femtosecond pulse (duration $\sim 10^{-15}$ s) is tightly focused inside the bulk of a glass substrate, the instantaneous intensity at the focal spot ($\sim 10^{13}$–10^{14} W/cm^2) becomes sufficient to trigger multiphoton and tunnel ionization processes [24]. Energy is deposited exclusively within the focal volume through these nonlinear mechanisms, leaving the surrounding material and the glass surface unaffected. This enables the inscription of arbitrary three-dimensional patterns deep inside the substrate simply by scanning the focal spot along the desired trajectory.

A key advantage of femtosecond pulses over longer-pulse or continuous-wave lasers is the absence of a significant heat-affected zone (HAZ). Because the pulse duration is shorter than the characteristic thermal diffusion time of the material ($\sim$ ps–ns), energy is deposited before heat can propagate to the surroundings [24, 25]. The process is therefore considered *cold machining*: it induces permanent structural modification at the focal point without melting, burr formation, or thermal cracking in the adjacent glass. Depending on the irradiation parameters—pulse energy, repetition rate, scan velocity, and numerical aperture—the nature of the modification can range from a smooth refractive index change, suitable for waveguide inscription, to the formation of self-organized nanoscale gratings (nanogratings), which are exploited in the FLICE process described below [26].

3.1.2 The FLICE Technique

Femtosecond Laser Irradiation followed by Chemical Etching (FLICE) is a two-step microfabrication process that combines the three-dimensional structuring capability of FLW with the material selectivity of wet chemical etching, allowing the fabrication of hollow microstructures buried inside a transparent substrate [27, 28].

In the first step, the femtosecond laser is scanned along the surfaces delimiting the desired hollow volume. Under appropriate irradiation conditions, the deposited energy induces the formation of self-oriented nanogratings within the focal volume [29]. These nanogratings consist of periodic nanoscale planes of modified material aligned perpendicularly to the laser polarization direction, and they significantly increase the chemical reactivity of the irradiated glass with respect to the surrounding unmodified bulk. In the second step, the substrate is immersed in an etchant solution—typically hydrofluoric acid (HF) or potassium hydroxide (KOH)—which dissolves the laser-modified material at a rate 100 times faster than the pristine glass [25]. The result is a smooth, hollow structure with the geometry defined by the laser scan, while the rest of the substrate remains intact and monolithic.

The orientation of the laser polarization plays a critical role in determining the etching efficiency. Because nanogratings align parallel to the laser polarization, orienting the polarization perpendicular to the longitudinal axis of an elongated channel results in nanogratings aligned along the channel direction, which favors acid penetration and increases the effective etching speed along that axis [14]. For complex multi-component geometries, the polarization direction must therefore be selected carefully for each sub-component in order to synchronize the etching rates across the entire structure and ensure complete and uniform material removal [14].

3.1.3 Application to Laser-Written Vapor Cells

The fabrication of the LWVC reported by Lucivero et al. [14] demonstrates the direct application of FLICE to the production of alkali-metal vapor cells in fused silica. The irradiation step was performed using a commercial femtosecond laser source emitting 300 fs pulses at 520 nm with a pulse energy of 410 nJ and a repetition rate of 1 MHz, focused inside a $10 \times 10 \times 5$ mm fused silica block using a $50\times$ microscope objective with numerical aperture 0.6 [14]. The irradiated geometry, displayed in Fig. 3.2, consisted of a cylindrical reservoir (5 mm diameter, 3.5 mm height) connected to a rectangular parallelepiped sensing channel (1×1 mm cross-section, 9.5 mm length).

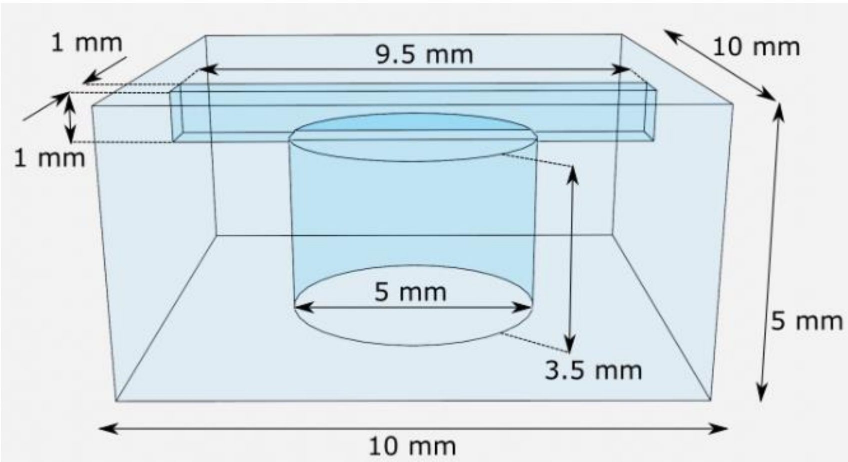

Figure 3.2: Laser-Written Vapor Cell manufactured using the FLICE technique. It consists of a cylindrical reservoir (5 mm diameter, 3.5 mm height) connected to a rectangular parallelepiped sensing channel (1×1 mm cross-section, 9.5 mm length). From [14]

Chemical etching was subsequently performed in a 20% aqueous HF solution at 35°C for approximately 18 hours, after which the sensing chamber was left buried at approximately 300 μm below the top surface of the substrate [14]. This stand-off distance could in principle be reduced to a few tens of micrometers, enabling near-field sensing geometries with sub-millimeter proximity to an external sample. The inner surfaces of the etched structure present a residual roughness in the range 50–100 nm RMS, which introduces optical scattering losses of approximately 8% per facet [30]. This roughness can be substantially reduced, down to 20 nm RMS, by performing a thermal annealing step after etching [31], which also partially relaxes residual stress in the glass.

The maskless and direct-write nature of FLICE makes it particularly well suited to rapid prototyping: new cell geometries can be implemented without any change of lithographic mask or process chemistry. Furthermore, since the entire cell is carved from a single monolithic substrate, there are no bonded interfaces inside the sensing volume, eliminating a potential source of outgassing or contamination. These properties, combined with the compatibility of fused silica with anti-relaxation coatings and with waveguide inscription by the same laser system, make FLICE the enabling technology for the LWVC platform [14, 26].

3.2 Optical Contact Bonding

Optical contact bonding is a glueless joining process in which two closely conformal surfaces adhere purely through intermolecular forces, without the need for adhesives or intermediate layers [32]. This property makes it particularly attractive for the hermetic sealing of Laser-Written Vapor Cells and other glass-based atomic sensors, where outgassing from adhesives, thermal incompatibility, or optical absorption at the bond interface can compromise device performance [33].

Intermolecular forces are, in isolation, insufficient to create a robust bond between two substrates. However, when mating surfaces conform to each other with nanometric precision, the resulting *intimate contact* over a sufficiently large area produces intermolecular interactions with a measurable macroscopic effect [34]. The bonding process can be understood through an energy balance: a bond front propagates across the interface if and only if the reduction in surface energy associated with the newly formed contact exceeds the elastic strain energy that must be stored in the wafers as they deform to accommodate each other's surface deviations [35, 36, 37]. For a surface gap of height h and lateral extent R on a wafer of thickness t and Young's modulus E, the bonding criterion takes the approximate form

$$\gamma \gtrsim C \, \frac{E \, t^3 \, h^2}{R^4}, \tag{3.1}$$

where γ is the work of adhesion per unit area and C is a geometry-dependent numerical prefactor [35, 36]. In practice, this imposes strict roughness and waviness requirements on both surfaces [32]. When these requirements are met and appropriate surface treatments are applied, bonding energies comparable to that of the bulk material ($2.5\,\mathrm{J\,m^{-2}}$) can be achieved [38].

Achieving such bond strengths requires a carefully sequenced preparation protocol. The following sections describe the key elements of this protocol: material selection and surface quality requirements, chemical treatment, plasma treatment, annealing, and mechanical pressure assistance. Each step builds on the previous one; together, they determine the final bond quality and make optical contact bonding a viable low-temperature process for LWVC fabrication [39].

3.2.1 Types of Glasses & Quality Requirements

The success of optical contact bonding critically depends on both the material properties of the glass substrates and their surface quality. Different glass types exhibit distinct chemical compositions, thermal characteristics, and mechanical behaviors that directly influence bonding performance. This section examines the most commonly employed glasses and establishes the surface quality requirements necessary for achieving robust, void-free bonds.

Fused silica and quartz (SiO_2, amorphous and crystalline, respectively) represent the gold standard for optical contact bonding. As characterized by Brückner [40], these materials offer ultra-high purity grades (>99.99% SiO_2) that minimize contamination and ensure uniform bonding behavior across the interface. Furthermore, with a low thermal expansion coefficient of approximately

$0.5 \times 10^{-6}\,\mathrm{K}^{-1}$, fused silica undergoes minimal dimensional changes, thereby significantly reducing thermal stress at the interface. These materials also provide excellent optical transmission from UV to IR wavelengths ($200\,\mathrm{nm}$ to $2500\,\mathrm{nm}$) [41]. Additionally, their chemical inertness permits aggressive cleaning treatments without substrate degradation, while their high damage threshold makes them suitable for high-power laser systems [42]. The primary distinction between the two lies in their structural organization: fused silica is amorphous and isotropic, while quartz is crystalline and exhibits optical anisotropy [40]. For most bonding applications—and in particular for LWVCs—synthetic fused silica is preferred due to its isotropic properties and the exceptional surface quality achievable through polishing [43].

Borosilicate glasses, particularly Borofloat 33 (81% SiO_2, 13% B_2O_3, 4% Na_2O/K_2O, 2% Al_2O_3), have gained widespread adoption in MEMS fabrication and vacuum cell manufacturing [44, 37]. They feature a moderate thermal expansion with a CTE of approximately $3.25 \times 10^{-6}\,\mathrm{K}^{-1}$ [45], which is highly compatible with silicon ($2.6 \times 10^{-6}\,\mathrm{K}^{-1}$), making this glass ideal for glass-to-silicon bonding [23]. Borosilicate glasses also maintain adequate chemical resistance to standard cleaning treatments, though they are less robust than fused silica [44]. Moreover, they are highly cost-effective and readily available in commercial wafer formats with polished surfaces that meet stringent bonding requirements.

Beyond these common types, several other specialized glass families address niche requirements. Aluminosilicate glasses offer higher mechanical strength, improved thermal shock resistance [46], and lower permeation rates for certain gases—such as helium—which is particularly relevant for LWVC hermeticity [47, 48]. Ultra-low-expansion glasses such as Zerodur, with CTE values near zero, are indispensable in applications demanding extreme dimensional stability, including space telescopes and lithography systems [49].

The bonding criterion established in Eq. (3.1) directly motivates strict surface quality requirements, which are summarized in Table 3.1. For a given material—with fixed E, t, and γ—the criterion sets an upper bound on the tolerable gap amplitude h as a function of its lateral scale R. In practice, three parameters must be strictly controlled. First, the surface root-mean-square (RMS) roughness must remain below $0.5\,\mathrm{nm}$ and the peak-to-valley roughness below 2 nm over the bonding area [23]. This guarantees intimate contact and allows intermolecular forces to dominate. At these length scales, the gap amplitude h is sufficiently small that even short lateral extents R satisfy Eq. (3.1). Furthermore, the bonding energy is proportional to the real area of contact, which decreases with increasing roughness independently of whether the surfaces close at the macroscale [50]. Second, flatness and waviness are equally critical. Total thickness variation (TTV) must remain below $1\,\mu\mathrm{m}$ for $100\,\mathrm{mm}$ diameter wafers, while surface flatness must be better than $\lambda/10$ (approximately $60\,\mathrm{nm}$ at $633\,\mathrm{nm}$) for precision optical applications. Intermediate-scale waviness is particularly detrimental because its lateral extent R is not large enough for the R^4 denominator in Eq. (3.1) to suppress the elastic energy cost, yet its amplitude h can reach tens of nanometres, far exceeding that of fine surface roughness [38, 51]. Finally, particulate contamination must be rigorously avoided. Bonding must be performed under cleanroom conditions (Class 100 or better, ISO 5), as even submicron particles can nucleate localized voids that propagate across the interface. Consequently, surface particle counts must be minimized, with particular attention given to particles larger than $0.1\,\mu\mathrm{m}$ [52].

Meeting these geometric requirements is necessary but not sufficient for bonding success. Even polished surfaces that satisfy all roughness and flatness criteria above retain organic contaminants, metallic impurities, and an insufficient density of reactive surface groups. Chemical treatment

Table 3.1: Surface quality requirements for optical contact bonding.

Parameter	Requirement	Reference
RMS roughness	$< 0.5\,\mathrm{nm}$	[23]
Peak-to-valley roughness	$< 2\,\mathrm{nm}$	[23]
TTV (100 mm wafer)	$< 1\,\mu\mathrm{m}$	[23]
Flatness (clear aperture)	$< \lambda/10 \approx 60\,\mathrm{nm}$	[38]
Min. particle size	$< 0.1\,\mu\mathrm{m}$	[52]
Cleanroom class	ISO 5 (Class 100)	[52]

addresses these molecular-scale obstacles, as described in the following section.

3.2.2 Chemical Treatment

Chemical treatment pursues two distinct but complementary objectives. The first is *cleaning*: removing organic matter, metallic ions, and particulate contamination that would prevent intimate contact regardless of how well the surface geometry has been prepared. The second is *activation*: maximising the density of reactive silanol groups (Si-OH) at the surface, which serve as the molecular precursors to the interfacial hydrogen bonds formed upon contact and the covalent siloxane bridges formed during annealing [53]. In practice, both objectives are addressed by the same sequence of chemical baths, but they operate through distinct mechanisms that are worth understanding separately.

Chemical Cleaning

Solvents. The cleaning sequence begins with mechanical degreasing using organic solvents. Acetone dissolves fats, waxes, and photoresist residues [54]. Ethanol follows as a miscible rinse: it removes residual acetone without leaving ionic residues, and its high volatility ensures rapid, streak-free evaporation. These steps reduce the organic load before the substrates enter the more aggressive chemical baths, which are expensive and whose lifetime is shortened by excessive organic contamination.

Piranha Solution (H_2SO_4:H_2O_2, 3:1). The primary wet-chemical agent for stubborn organic removal is Piranha solution, a strongly exothermic mixture that operates through two concurrent mechanisms at different rates. The first and faster mechanism is dehydration: concentrated H_2SO_4 strips hydrogen and oxygen from organic molecules as water, exploiting the highly favourable enthalpy of hydration ($\Delta H \approx -880\,\mathrm{kJ\,mol^{-1}}$). This process carbonises organic residues within seconds, reducing them to water-soluble fragments [55]. The second, slower mechanism is oxidation: as described in Figure 3.3, H_2SO_4 reacts with H_2O_2 to form Caro's acid (H_2SO_5), a potent oxidising agent that generates hydroxyl radicals capable of breaking C–H and Si–O bonds alike [56]. The net effect on the glass surface is twofold: organic contaminants are eliminated, and the oxidising radicals attack the outermost Si–O–Si (siloxane) bonds, cleaving them and leaving the exposed silicon atoms terminated as Si-OH groups. Piranha therefore acts simultaneously as a cleaner and as a hydroxylating agent, and substrates emerge from the bath with a highly hydrophilic, silanol-rich surface, such as the one displayed in Figure 3.4.

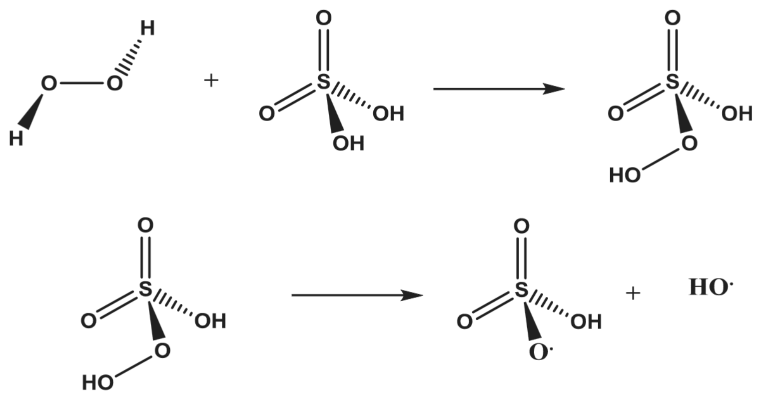

Figure 3.3: Hydroxyl radicals formation upon the decomposition of Caro's acid in Piranha solution. From [57].

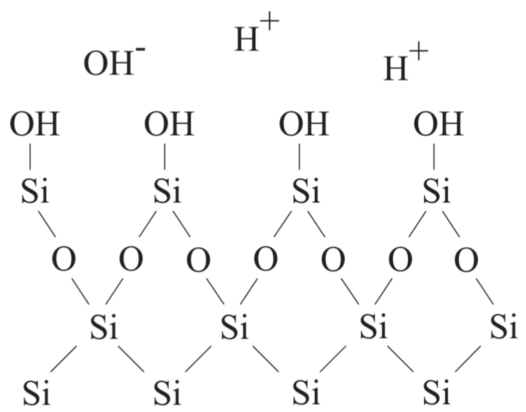

Figure 3.4: Schematic representation of a hydroxylated/hydrophilic surface. From [56].

RCA-1 (NH_4OH**:**H_2O_2**:**H_2O**, 1:1:5 at** 70–80 °C**).** The RCA Standard Clean 1 (SC-1), introduced by Kern and Puotinen in 1970 [58], targets light organic films and, most importantly, sub-micron particulate contamination. Its mechanism relies on a dynamic oxide dissolution–regeneration cycle: in the alkaline environment maintained by NH_4OH, the H_2O_2 simultaneously oxidises the outermost silica layer to form a thin chemical oxide, while the NH_4OH dissolves this oxide at approximately the same rate [56]. This cycle continuously regenerates the surface, and the key consequence for particle removal is mechanical: as the oxide beneath an adhered particle is dissolved away, the particle becomes undercut and loses its mechanical anchorage to the substrate, leaving it free to diffuse into solution. At the basic pH of SC-1, both the oxidised glass surface and the surfaces of most particle contaminants acquire a negative Zeta potential (as displayed in Figure 3.5), creating an electrostatic repulsion that prevents re-deposition once particles are dislodged [56]. The repeated oxidation–dissolution cycles also leave the surface with a dense, freshly formed termination of Si-OH groups, contributing to surface activation. Additionally, the NH_4OH also serves to remove some metals such as Cu, Au, Ag, Zn, and Cd, and some elements from other groups such as Ni, Co, and Cr [56].

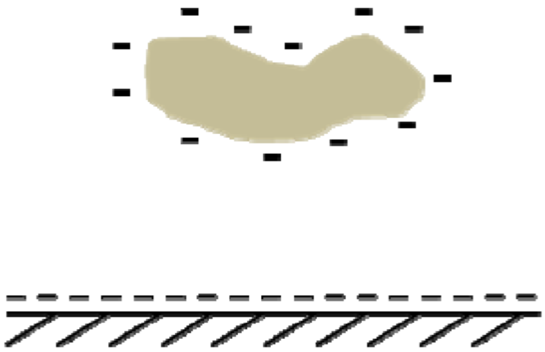

Figure 3.5: Schematic representation of a contaminant being lifted from a quartz surface due to the Zeta potential. Adapted from [59].

RCA-2 (HCl**:**H_2O_2**:**H_2O**, 1:1:6 at** 70 °C**).** The second RCA step (SC-2) targets metallic contamination introduced either during previous processing or by the SC-1 bath itself, which can deposit trace metal ions onto the surface. Its mechanism is straightforward: H_2O_2 oxidises metallic species to their ionic forms, while HCl provides a strongly acidic environment in which these ions form soluble chloride complexes that are rinsed away with the bath [58, 56]. The acidic pH also reverses the Zeta potential of the surface, which is advantageous for removing metal hydroxide colloids that might have survived SC-1. SC-2 has no significant activation effect on the silanol density.

Ultrasonic Agitation. For all bath steps, ultrasonic agitation substantially enhances contaminant removal. High-frequency sound waves generate microscopic cavitation bubbles in the liquid; their violent implosion near the glass surface produces localised pressure jets that dislodge particles from recesses and pores that are otherwise inaccessible to chemical soaking alone [60].

Chemical Activation

The cleaning steps described above increase surface hydrophilicity as a side effect of their primary cleaning action. Dedicated activation takes this further. A hydroxylated silica surface exposed to ambient humidity rapidly adsorbs water molecules onto its Si-OH sites, building up a chemisorbed water layer one to a few monolayers thick [61]. When two such surfaces are brought into contact, this water layer bridges the opposing silanol groups through a network of hydrogen bonds [53]:

$$\equiv\text{Si-OH}\cdots\text{H}_2\text{O}\cdots\text{HO-Si}\equiv \tag{3.2}$$

This water-mediated hydrogen-bonding network provides the initial room-temperature adhesion required to handle the bonded pair. As the interface is annealed at temperatures above ~100 °C, the water is gradually expelled and adjacent silanol pairs condense directly into covalent siloxane bridges [62, 63]:

$$\equiv\text{Si-OH} + \text{HO-Si}\equiv \xrightarrow{T>100\,^\circ\text{C}} \equiv\text{Si-O-Si}\equiv\ + \text{H}_2\text{O} \tag{3.3}$$

The bond energy after annealing scales with the number of siloxane bridges formed per unit area, which in turn is limited by the silanol density present at the time of contact [62, 61]. Since no treatment after bonding can introduce new silanol groups at the buried interface, the activation state achieved before contact is an upper bound on the final bond strength.

The most widely used activation step is an additional SC-1 soak applied immediately before bonding. The dissolution–regeneration cycle described above continuously exposes fresh, uncontaminated silica with a maximum density of Si-OH groups estimated at approximately $4.6\,\mathrm{nm}^{-2}$ on a fully hydroxylated amorphous silica surface [61]. In practice, a water contact angle below 5° after the final DI water rinse is used as the operational criterion for successful activation, as it indicates that the silanol density is sufficient for spontaneous bonding upon contact [23, 37]. An alternative activation method is exposing the substrates to a *plasma treatment*, as explained in Section 3.2.3.

3.2.3 Plasma Treatment

Plasma treatment unifies the cleaning and activation functions described above into a single process, while adding a third benefit unavailable through wet chemistry alone: the enhancement of interfacial water and gas diffusivity. Together, these mechanisms make plasma treatment one of the most impactful steps in the bonding protocol, capable of yielding high bonding energies even at low annealing temperatures. In a plasma cleaner, a mixture of electrons, ions, free radicals, and photons acts simultaneously on the glass surface [32, 64, 65]. This triggers a wide range of physical and chemical effects that fundamentally alter the substrate's surface, as illustrated in Figure 3.6.

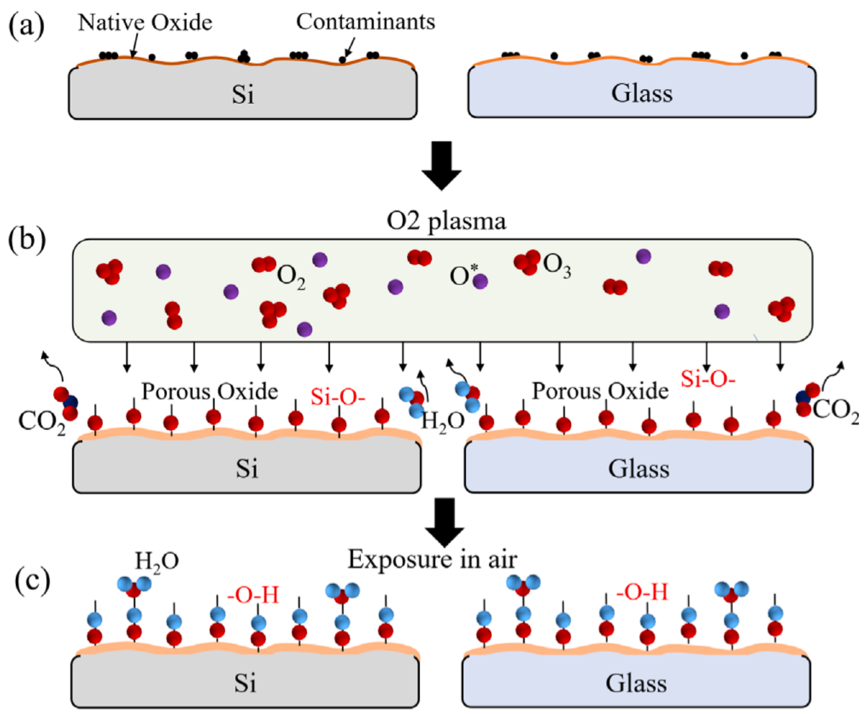

Figure 3.6: Effects of plasma treatment on silica surfaces. Plasma removes contaminants and increases the density of available silanol groups, while generating a thin less dense, nanoporous subsurface layer, increasing the diffusivity of water and trapped gases at the interface. Adapted from [66].

Chemically, the plasma decomposes and volatilizes organic contaminants through physical bombardment and chemical reactions, significantly increasing the fraction of clean, bondable area [67, 68]. More importantly, the energetic bombardment breaks stable surface siloxane rings (Si-O-Si), creating a high density of highly reactive dangling bonds (Si* and O*). Upon subsequent exposure to ambient moisture or a deionized water (DIW) rinse, these highly energetic sites rapidly hydrate to form dense silanol (Si-OH) terminations. This renders the surface superhydrophilic—typically reducing the water contact angle to below 5°—and provides the essential chemical foundation for strong covalent bonding [36, 37].

Physically, the plasma ion bombardment alters the structure of the top few nanometers of the silica substrate, generating a slightly less dense, nanoporous subsurface layer. This modified, open network is crucial because it acts as a reservoir, significantly increasing the diffusivity of water and trapped gases at the interface [69, 70]. During the annealing phase, the rapid evacuation of interfacial water shifts the chemical equilibrium to promote the conversion of weak, reversible hydrogen bonds into stronger Si-O-Si covalent bonds. Simultaneously, the absorption of gases into this porous layer prevents the macroscopic bubble (void) formation that often plagues non-activated bonded pairs.

To achieve these optimal surface conditions, process gases such as O_2, N_2, and Ar are commonly used [65, 68], and chamber parameters must be carefully tuned. The choice of gas dictates the dominant activation mechanism: O_2 plasma is highly reactive, effectively oxidizing organic residues and optimizing the surface for silanol formation, whereas Ar plasma relies entirely on physical sputtering to displace surface atoms. Sequential treatments or gas mixtures are often employed to synergize the physical cleaning power of Ar with the chemical activation of O_2 [64, 65]. Furthermore, plasma power and exposure time govern the energy dose delivered to the substrate. While sufficient energy is required to break the siloxane bonds, excessive power or prolonged exposure beyond a few minutes induces severe physical sputtering. This degrades the surface by increasing its root-mean-square (RMS) roughness, which must generally remain below 1 nm to avoid unbonded macroscopic areas [32]. Consequently, short, low-to-medium power treatments are preferred.

An example of how the surface reactivity evolves with plasma treatment time is illustrated in Figure 3.7. As shown, the surface energy exhibits a distinct peak at an optimal exposure of 60 s. Beyond this threshold, additional plasma exposure leads to a gradual degradation of the surface energy, a phenomenon primarily attributed to surface overetching and the consequent increase in roughness.

Chamber pressure also plays a vital role, as it determines the mean free path of the plasma species; lower pressures increase ion kinetic energy and sputtering risk, whereas higher pressures favor a higher density of lower-energy radicals for gentler chemical modification.

A critical procedural constraint is that plasma treatment must always follow all wet chemical cleaning or activation steps, as subsequent exposure to chemicals can etch the newly activated surface and negate the plasma's benefits [71]. Additionally, plasma-activated surfaces are subject to a "hydrophobic recovery" effect, meaning they lose their reactivity if left in ambient air for extended periods due to the re-adsorption of airborne contaminants and surface reconstruction. Therefore, bonding should occur shortly after activation. As mentioned, a brief DIW rinse immediately after the plasma step is highly beneficial to ensure maximum silanol conversion [66]. The recommended workflow is therefore: (1) chemical cleaning, (2) chemical activation (e.g., RCA-1), (3) plasma treatment, (4) DIW rinse, and (5) immediate pre-bonding and annealing.

Ultimately, plasma treatment is especially valuable for low-temperature optical contact bonding in

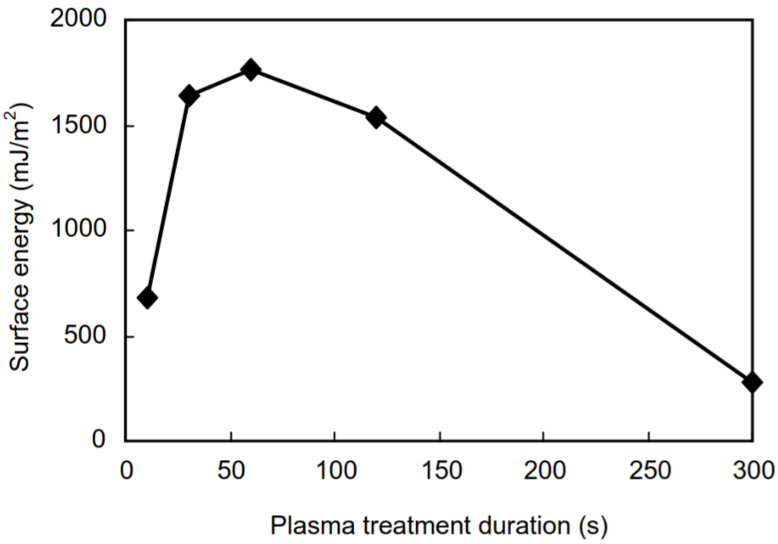

Figure 3.7: Bonding energy as a function of plasma treatment duration, highlighting the transition from optimal surface activation to degradation via overetching after 60 s **of exposure. Adapted from [39].**

LWVCs. It produces bond energies exceeding $1\,\mathrm{J\,m^{-2}}$ at annealing temperatures as low as $100\,°\mathrm{C}$ [63, 66], which is well below the thermal thresholds that would damage or deform pre-structured glass components.

3.2.4 Thermal Treatment

When two sufficiently flat and clean glass surfaces are brought into contact, the bonding process initiates through the formation of hydrogen bonds between opposing silanol groups [53]. These bonds can subsequently be converted into permanent covalent siloxane bonds via the condensation reaction:

$$\equiv\text{Si-OH} + \text{HO-Si}\equiv \xrightarrow{T > 100\,°\text{C}} \equiv\text{Si-O-Si}\equiv \; + \text{H}_2\text{O} \tag{3.4}$$

For this reaction to proceed irreversibly, water must be effectively removed from the interfacial network [32]. This is accomplished through *annealing*, which involves baking the bonded assembly in a strictly controlled thermal environment governed by three primary parameters: the temperature ramp, the equilibrium temperature, and the total annealing time.

The *temperature ramp* (heating and cooling rate) must be kept as low as possible, typically restricted to fractions of a degree per minute. Rapid temperature alterations induce severe transient thermal gradients and differential stresses within the glass substrates. These thermal stresses can compromise the mechanical integrity of the assembly, degrade optical performance via stress-induced birefringence, or cause catastrophic delamination and structural cracking due to thermal shock [72].

In contrast, the *equilibrium temperature* directly controls the kinetics of water diffusivity at the interface. Elevated temperatures accelerate moisture evacuation and promote localized viscous flow, which fundamentally increases atomic-scale surface conformity [32]. Generally, the interfacial surface energy increases progressively with higher annealing temperatures, eventually reaching a plateau. However, in low-to-moderate temperature regimes, this plateau does not reflect the bulk fracture energy of the material; instead, it represents a state of equilibrium determined by the maximum density of silanol-to-siloxano conversions achievable under the restricted thermal budget and the remaining interfacial moisture. Achieving true bulk-like bonding energies without plasma would require high thermal budgets (typically above $600\,^{\circ}\mathrm{C}$) that are strictly incompatible with precision devices incorporating thin-film optical coatings, pre-structured channels, or integrated electronics [39]. In these specialized scenarios—including LWVCs, where laser-written microstructures must preserve their exact structural dimensions—the integration of plasma activation becomes indispensable. Plasma treatment raises the initial surface energy and subsurface water diffusivity sufficiently to maximize the covalent bond density at temperatures as low as $100\,^{\circ}\mathrm{C}$ [67], a strategy known as *Low-Temperature Direct Bonding*. The characteristic evolution of bonding energy within this optimized thermal framework is illustrated in Figure 3.8.

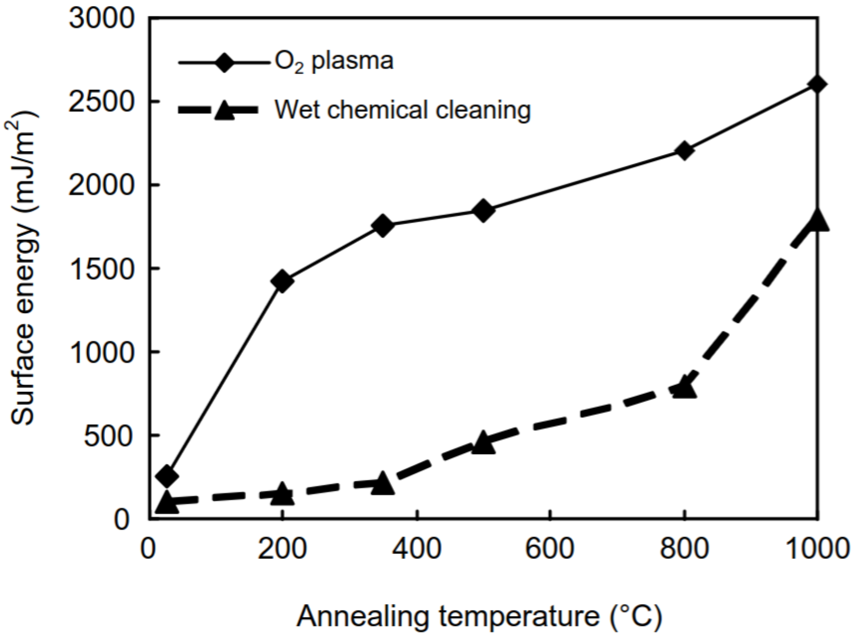

Figure 3.8: Evolution of bonding energy as a function of annealing temperature. Adapted from [39].

Regarding the *annealing time*, the bond energy increases monotonically following a predictable logarithmic trend, a behavior observed during both dedicated thermal baking and prolonged storage at room temperature [67]:

$$\gamma \propto \ln\left(\frac{t}{\mu}\right), \tag{3.5}$$

where t represents the elapsed annealing time and μ is a temperature-dependent characteristic time constant. Thermal annealing effectively reduces μ by several orders of magnitude, accelerating the convergence toward maximum bond strength, governed by the surface silanol density. Consequently, typical annealing durations required to stabilize high-performance optical components range from 2 to 24 hours [67, 66, 39, 37, 73], as detailed in Figure 3.9.

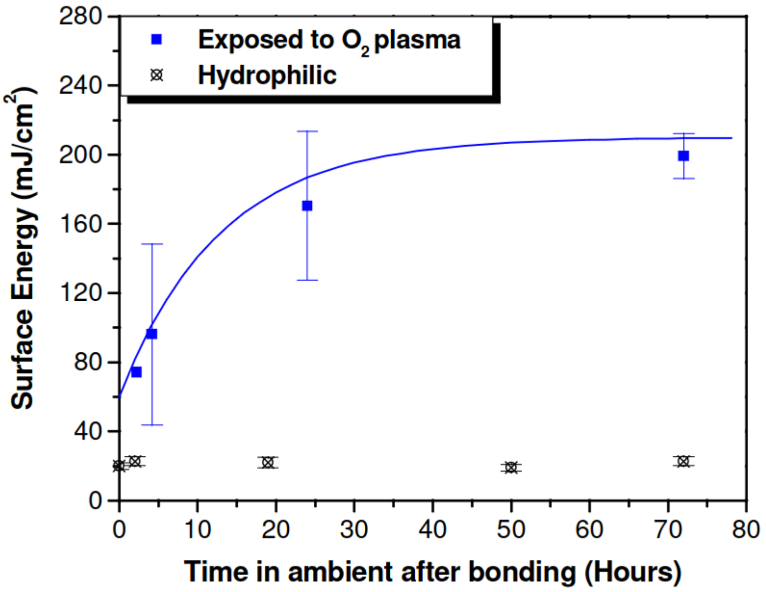

Figure 3.9: Evolution of bonding energy with respect to annealing time. The bond strength follows a logarithmic growth law, where thermal annealing reduces the characteristic time constant μ to accelerate convergence toward maximum bond strength. From [67].

3.2.5 Pressure Treatment

Applied mechanical pressure during the bonding phase can promote intimate contact by elastically deforming surface waviness and closing residual macroscopic gaps at the interface [74, 73], as illustrated in Figure 3.10. This mechanical assistance is primarily beneficial for substrates with insufficient flatness, where the spontaneous van der Waals attraction alone cannot provide enough energy to pull the surfaces together [35, 38]. In such non-ideal scenarios, the bonding condition is governed by a strict energy balance: the bonding wave propagates only if the reduction in surface energy associated with the newly formed interface exceeds the elastic strain energy stored in the mechanically deformed glass. External pressure temporarily shifts this balance in favor of bonding by forcing asperities into intimate contact [35, 51].

For well-polished, optically flat wafers, the role of external pressure is not only ambiguous but potentially detrimental. Since the bonding energy at room temperature is governed primarily by the real area of contact—which is dictated by surface roughness at the nanoscale rather than by externally applied load—additional mechanical pressure does not intrinsically increase the bond energy [50, 75]. In this regime, the spontaneous intermolecular forces are already sufficient to achieve full intimate contact. Crucially, applying unnecessary pressure to seemingly flat wafers severely amplifies the damage caused by microscopic particulate contamination. A single sub-micron particle trapped at the interface under high external load acts as a point-stress concentrator. This generates severe Hertzian contact stresses that can easily exceed the fracture toughness of fused silica, inducing subsurface micro-cracking, permanent local deformation, and the formation of macroscopic unbonded areas (Newton rings) [32, 36].

Consequently, pressure magnitude must be optimized with extreme caution. Research on Borofloat

33 wafers demonstrates that a modest load of just 0.07 MPa maximizes the flexural strength of the final assembly; applying pressures beyond this strict threshold introduces stress concentrations that cause premature mechanical failure [73]. Furthermore, non-uniform pressure distributions pose significant risks to device functionality. Asymmetric loading induces stress birefringence and permanent wavefront distortion, which can critically compromise optical performance in precision devices such as atomic vapor cells [76, 77]. High-precision applications therefore require well-planarized pressure plates with peak-to-valley flatness on the order of a few micrometers, strictly fabricated from highly stiff materials like technical ceramics to prevent plate bending [78].

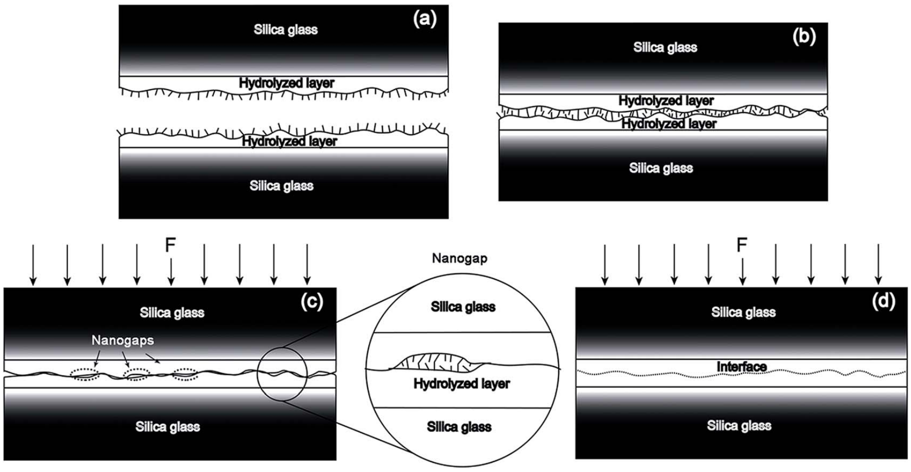

Figure 3.10: Effect of applied pressure on the bonding process. Forces perpendicular to the interface suppress void formation by closing residual gaps between surface asperities. From [74].

Beyond the magnitude of the load, applying a uniform, flat pressure across the entire wafer surface simultaneously is often counterproductive, as it seals the perimeter of the substrates prematurely and traps residual atmospheric gases in the center, forming large voids. To resolve this, Birckigt et al. [78] describe an optimized setup employing convex and concave pressure plates with carefully controlled waviness profiles of 5 µm and 3 µm, respectively. As shown in Figure 3.11, this engineered curvature forces the initial contact to occur at the geometric center of the wafers. The applied load then drives the contact front to propagate radially outward, actively squeezing out interfacial air and mitigating the risk of void entrapment. This insight underscores a fundamental principle of optical contact bonding: the geometric control of the contact-front dynamics is vastly more critical to achieving a void-free interface than the raw magnitude of the applied mechanical pressure.

3.2.6 Step-by-Step Experimental Procedure

3.3 Post-Treatment

3.3.1 Enhanced Helium Hermeticity

Helium plays a critical role in the operation of various atomic sensors, serving primarily as a working medium or buffer gas in high-sensitivity applications (e.g., magnetoencephalography [79]).

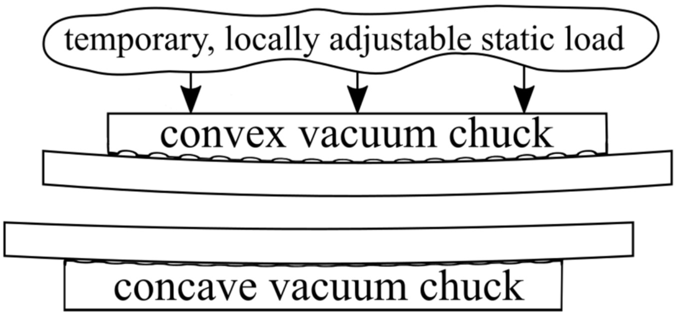

Figure 3.11: Pressure plate geometry described by Birckigt et al. [78]. The curved profiles guide contact front propagation outward from the center, reducing the probability of interface void formation.

However, due to its extremely small atomic radius, helium exhibits one of the highest permeation rates through glass walls compared to other gases and alkali metals typically confined within microelectromechanical systems (MEMS) vapor cells.

For high-precision applications where long-term cell hermeticity is paramount, aluminum oxide (Al_2O_3, or alumina) has been identified as a highly effective barrier [47, 48]. Researchers have found that utilizing aluminosilicate glass—a glass matrix enriched with Al_2O_3—or depositing thin Al_2O_3 coatings directly onto standard glass cells significantly mitigates helium diffusion.

As illustrated in Figure 3.13, the permeability constant (K) of aluminosilicate glass is between two and three orders of magnitude lower than that of conventional borosilicate glass. Furthermore, applying an Al_2O_3 coating to borosilicate cells can independently reduce helium permeability by one to two orders of magnitude. Synergistically combining an aluminosilicate glass substrate with the aforementioned Al_2O_3 coating yields the best results, achieving ultra-low permeabilities on the order of 10^{-22} $\mathrm{m^2s^{-1}Pa^{-1}}$.

3.3.2 Activation

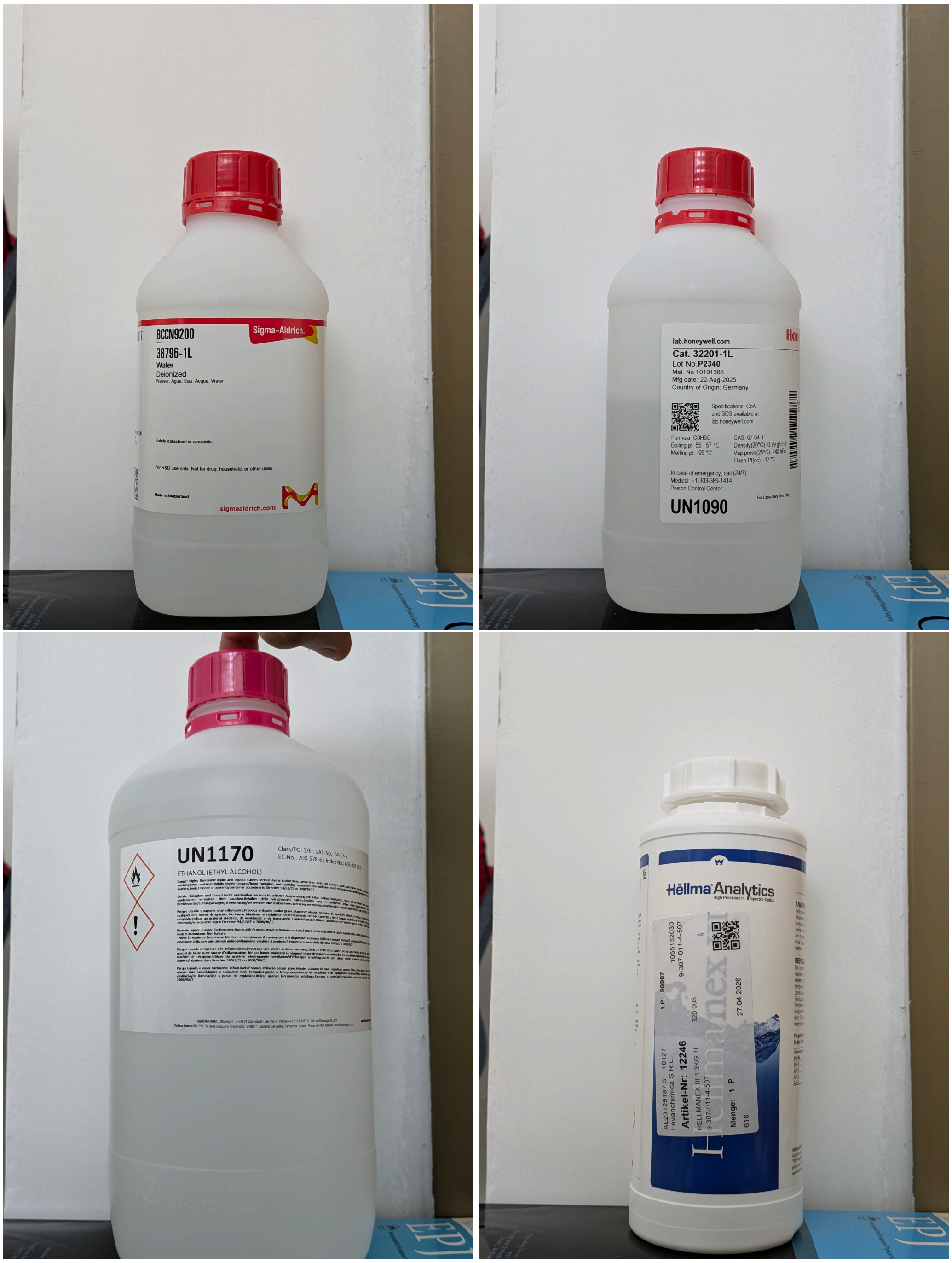

Figure 3.12: Chemicals employed in the experiment. (1) Deionized Water (DIW) (2) Acetone (3) Ethanol (4) Hellmanex III soap

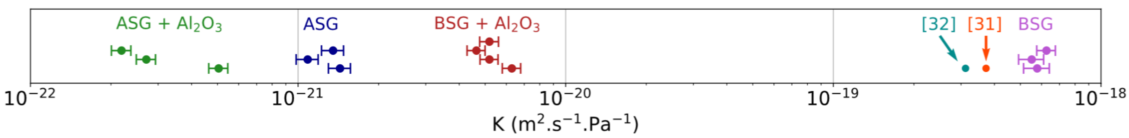

Figure 3.13: Comparison of the permeability constant K across uncoated borofloat, uncoated aluminosilicate, Al_2O_3-coated borofloat, and Al_2O_3-coated aluminosilicate glasses. From [47].

Chapter 4

Characterization and Quality Control

Ensuring excellent quality standards throughout the fabrication process is a fundamental requirement in the development of atomic sensors based on alkali vapor cells. The performance of such devices depends critically on several interdependent physical properties of the fabricated components, ranging from the nanometric quality of the glass surfaces to the mechanical integrity of the bonded interfaces and the long-term hermeticity of the sealed cell. Consequently, a rigorous characterization and quality-control framework must accompany each stage of the fabrication pipeline, from the initial substrate preparation through to the final sealed device.

Pre-bonding Characterization

Before any bonding attempt is made, the substrate surfaces must be thoroughly characterized to ensure they meet the stringent requirements of low-temperature bonding techniques such as Optical Contact Bonding. Two complementary methods are employed at this stage. First, Atomic Force Microscopy (AFM) is used to inspect the surface topography at the sub-nanometric scale, yielding quantitative roughness parameters over representative areas of the sample. This allows both the as-received substrate quality to be verified against manufacturer specifications and the effect of surface treatments—such as cleaning or plasma activation—to be assessed. Second, contact angle measurements provide a direct and experimentally simple assessment of surface hydrophilicity: a low water contact angle indicates a highly hydroxylated surface, which is a prerequisite for strong and defect-free hydrophilic bonding. Together, these two techniques establish whether a substrate is ready for bonding, and provide a baseline against which post-process surfaces can be compared.

Post-bonding Characterization

Once bonding has been performed, the focus shifts to evaluating the integrity of the newly formed interface and, ultimately, of the complete sealed device. As a first non-destructive step, visual inspection based on thin-film interference (Newton's rings) reveals the presence of macroscopic defects, trapped particles, or surface flatness deviations at the bonding interface, providing immediate qualitative feedback on the bonding outcome. When a quantitative assessment is required, two complementary mechanical tests are employed: the *Double Cantilever Beam* (DCB) method, also known as the razor blade or blade insertion test, which quantifies the interfacial energy release rate G by measuring crack propagation length; and the tensile (pull) test, which directly measures the bonding strength in terms of the critical stress σ_C required to fracture the interface.

Finally, once the cell has been sealed and activated with the alkali metal vapor, the stability and purity of the internal gas environment must be carefully monitored over time. Any compromise in hermeticity—allowing atmospheric gases to permeate the sealed cavity—would degrade sensor performance through unwanted pressure broadening and frequency shifts of the atomic transitions. Laser absorption spectroscopy, exploiting the pressure dependence of the Lorentzian linewidth and line-center shift of the alkali D-line transitions, provides a non-invasive and highly sensitive method for quantifying the internal gas pressure and for establishing meaningful upper bounds on the leak rate.

This section presents in detail the theoretical framework and experimental procedures underlying each of these characterization techniques, and reports the results obtained during the quality assessment of the vapor cells fabricated in this work.

4.1 Surface inspection using an Atomic Force Microscope

Atomic Force Microscopy (AFM) is a sophisticated type of scanning probe microscopy, which consists of a physical probe that scans the specimen in order to produce images of surfaces. Atomic Force Microscopes can map the topology of a surface with sub-nanometric resolution, which makes them an ideal tool for surface inspection in the context of atomic sensors, and more specifically in the context of Optical Contact Bonding, due to the strict glass roughness requirements of this method. The experimental characterization presented in this work was carried out using the facili-

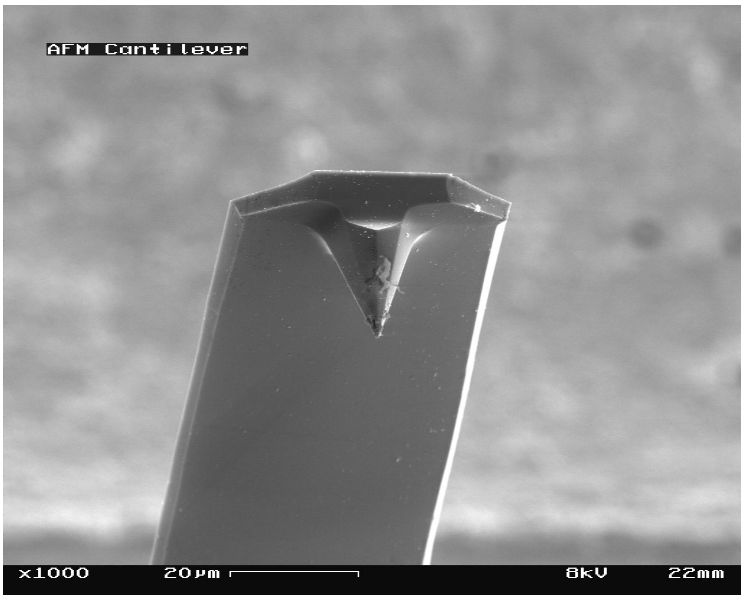

Figure 4.1: Example of the probe of an Atomic Force Microscope. From [80].

ties of the Physics Department at the *Politecnico di Bari* (POLIBA) in Bari, Italy.

A brand-new Nanoquartz glass disk, with a specified Root Mean Square (RMS) roughness of 500 pm and a Total Thickness Variation (TTV) of 2 µm, was inspected using a Park NX10 AFM. The measurements were performed with a Park Scout 350 tip in *tapping* mode (resonance frequency 320 kHz, scan rate 0.5 Hz). We examined the sample in three different spots of its surface to ensure consistency. The obtained data was subsequently analyzed using the *Gwyddion* software,

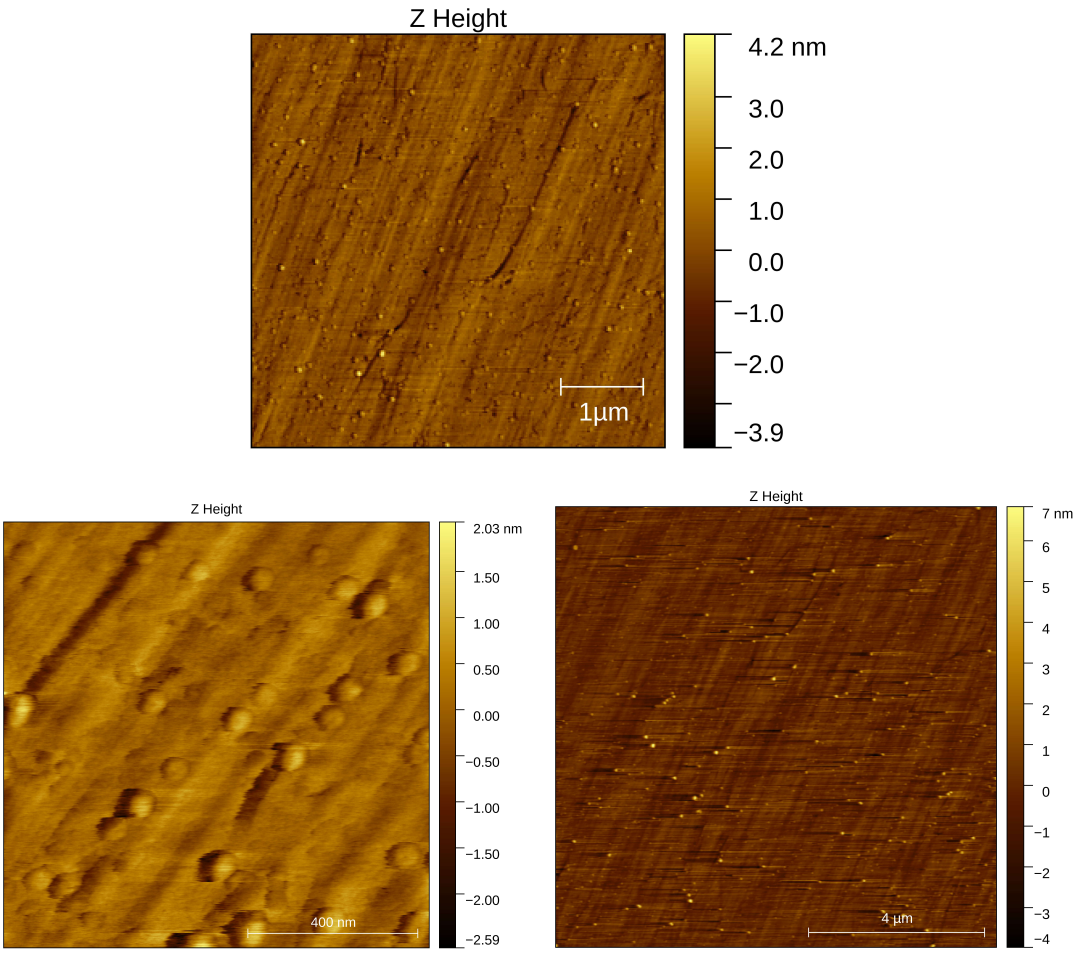

Figure 4.2: 2D Surface images obtained through AFM in three different parts of the pristine Nanoquartz disk's surface.

through which we obtained the two-dimensional (Figure 4.2) and three-dimensional (Figure 4.3) images. The resulting density distribution of roughness is displayed in Figure 4.4, and the detailed list of the surface parameters estimated from our measurements can be found in Table 4.1. The precise definition of these parameters is provided in [81].

The results confirm that the Root Mean Square roughness of the disk is around 500 pm, as suggested by the manufacturer. An alternative estimation of the surface roughness, known as *Mean roughness* (Sa) was found to be in the same order of magnitude. Additionally, it was determined that the total height (S_q) is in the range of a few nanometers. The mean height is approximately zero, suggesting that the surface's defects are uniformly distributed. The outstanding surface parameters observed in this disk comply with Optical Contact Bonding requirements and therefore it is suitable for the fabrication of Laser-Written Vapor Cells and high-precission atomic sensors.

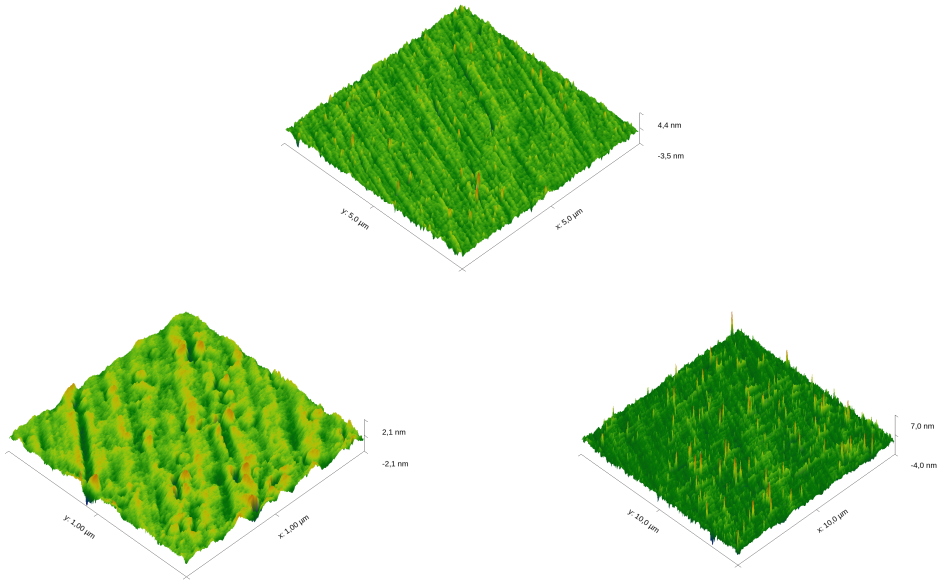

Figure 4.3: 3D Surface images obtained through AFM in three different parts of the pristine Nanoquartz disk's surface.

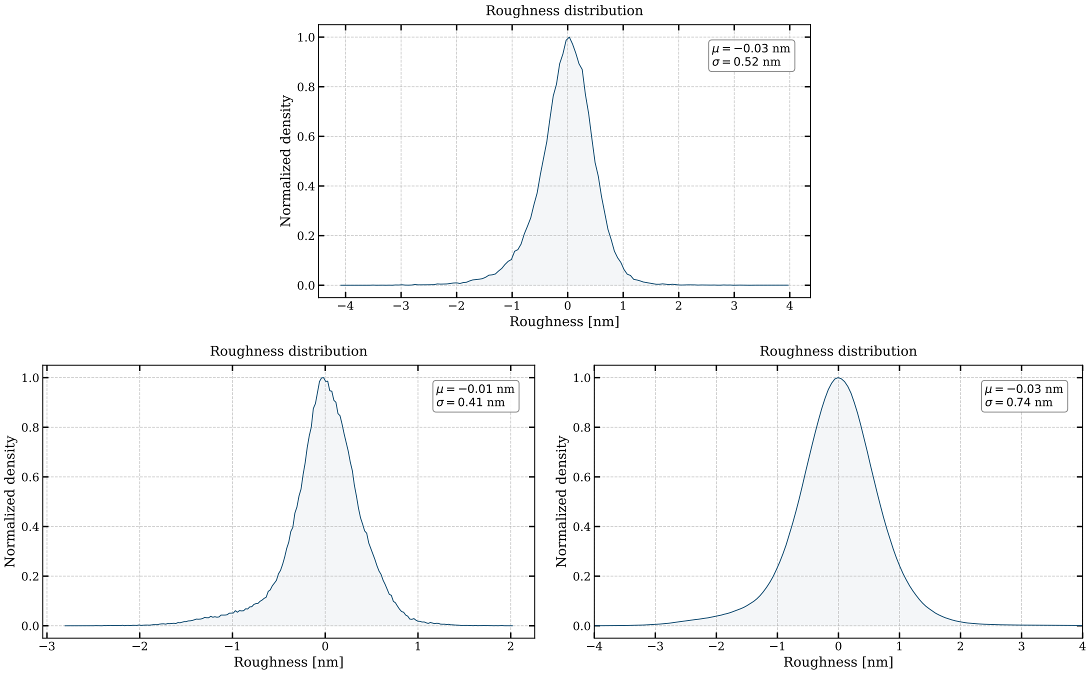

Figure 4.4: Surface roughness distribution obtained through AFM. It can be seen that most of the defects are below the 1 nm range. Furthermore, the mean (μ) is close to zero, confirming that the surface is uniform.

Sample	S_q [pm]	S_a [pm]	S_p [nm]	S_v [nm]	S_z [nm]	Mean height [nm]
1	518	387	4.02	-4.14	8.17	0.03
2	411	298	2.04	2.83	4.87	0.02
3	726	529	7.66	4.05	11.71	0.02

Table 4.1: Surface parameters obtained through AFM in three different parts of the surface: ***Root mean square roughness*** **(S_q),** ***Mean roughness*** **(S_a),** ***Maximum peak height*** **(S_p),** ***Minimum valley depth*** **(S_v),** ***Maximum height*** **(S_z) and** ***Mean height***

4.2 Contact Angle Measurement

The water contact angle (WCA) is the standard characterization method used in the optical contact bonding literature to assess the chemical state of a glass surface prior to bonding [23, 37]. In this simple method, the measured angle is a direct proxy for the density of reactive surface groups that govern bond formation.

The contact angle θ_C is geometrically defined as the angle formed at the three-phase boundary where a liquid droplet, the surrounding gas, and the solid substrate meet simultaneously, as illustrated in Figure 4.5. At mechanical equilibrium, the angle satisfies Young's equation [82]:

$$\cos\theta_C = \frac{\gamma_{sg} - \gamma_{sl}}{\gamma_{lg}}, \tag{4.1}$$

where γ_{sg}, γ_{sl}, and γ_{lg} denote the solid–gas, solid–liquid, and liquid–gas interfacial energies, respectively. This equation expresses the condition that minimises the total free energy of the system for a virtual displacement of the contact line [82]: the contact angle is determined entirely by the balance between the tendency of the liquid to spread over the solid (governed by $\gamma_{sg} - \gamma_{sl}$) and the tendency of the liquid–gas interface to contract (governed by γ_{lg}). As γ_{sg} increases relative to γ_{sl}, the liquid spreads more, and θ_C decreases.

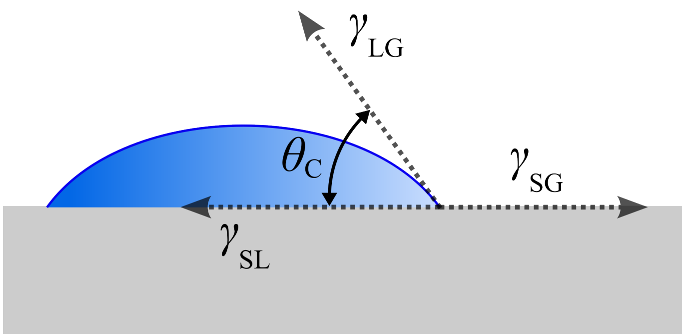

Figure 4.5: Geometrical definition of the contact angle θ_C at the three-phase boundary between a liquid droplet, the ambient gas, and the solid substrate. The equilibrium value is set by the balance of interfacial energies expressed in Young's equation (Eq. (4.1)). From [83].

On silica and glass surfaces, the magnitude of γ_{sg} is controlled by the density of surface hydroxyl groups, commonly referred to as silanol groups (Si-OH). Silanol groups are simultaneously hydrogen-bond donors and acceptors [61], which gives them a strong chemical affinity for water.

A fully hydroxylated silica surface therefore exhibits a very high γ_{sg} and a low γ_{sl}, resulting in low contact angles. As the silanol density decreases, whether due to thermal treatment, which converts Si-OH groups into hydrophobic siloxane bridges (Si-O-Si), or due to organic contamination, which physically blocks hydrophilic sites, γ_{sg} falls and γ_{sl} rises. By Young's equation, this directly translates into an increase in θ_C. The WCA thus provides an experimentally accessible measurement of the silanol surface density, without requiring the use of more complex or expensive techniques.

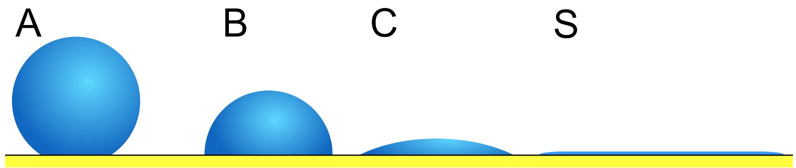

Figure 4.6: Representative contact angle regimes across different wettability states. Surface A ($\theta_C > 90°$) is hydrophobic; surfaces B, C, and D are sorted from partially to fully hydrophilic ($\theta_C \to 0°$). From [84].

The connection between the WCA and optical contact bonding is therefore direct: the silanol groups whose density is encoded in the contact angle are the same reactive sites that initiate bond formation. A low WCA is therefore not merely a qualitative indicator of surface cleanliness, but a quantitative predictor of the maximum achievable bond energy. For this reason, a contact angle below $\sim 5°$ is consistently used as the pass/fail threshold for surface activation in the direct bonding literature [23, 37].

4.3 Visual Inspection

Visual inspection is a primary, non-destructive method for evaluating the integrity of optical contact bonds. The physical principle underlying this technique is thin-film interference, which manifests visibly as Newton's rings when the contacting surfaces are illuminated.

In an ideal optical contact bonding process, the interfacial air gap is completely eliminated. Optically, this results in the disappearance of interference fringes, indicating a defect-free and continuous bond. However, the presence of Newton's rings during or after the bonding process serves as a direct indicator of interfacial defects. These interference patterns occur due to localized variations in the gap thickness between the two surfaces. Issues can be identified and evaluated through this phenomenon are trapped air and voids, particulate contamination and surface flatness deviations [85, 32]. Figure 4.7 shows an example of Newton rings.

Depending on the optical properties of the materials involved, visual inspection is conducted using different setups. For transparent optical components, monochromatic or white light interferometry is typically used. Conversely, for materials that are opaque in the visible spectrum, such as silicon wafers, infrared (IR) transmission imaging is routinely employed to visualize these interference patterns at the bonding interface [64]. Ultimately, an interfacial region completely devoid of visible Newton's rings serves as the standard criterion for a successful, defect-free optical contact bond.

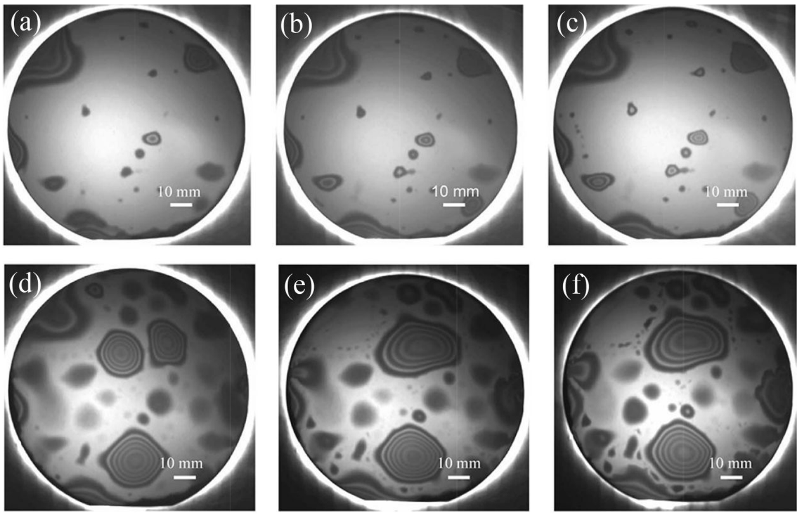

Figure 4.7: Examples of Newton rings formed in the interphase of Si/Si wafers. From [86].

4.4 Bonding Energy Test

ESTA SECCIO HA QUEDAT UN POC RARA The most widely employed technique for measuring wafer bonding strength is the *double cantilever beam* (DCB) method, also known as the blade insertion test, crack opening test, or Maszara's method [87]. It consists of inserting a thin blade or razor (typically 0.1 mm thick) into the bonding interface, which induces crack propagation along the bonded wafer pair [87]. This setup is illustrated in Figures 4.8 and 4.9. The bonding strength is quantified in terms of the energy release rate G (in J m $^{-2}$), which represents the surface energy required to create new surfaces during crack propagation. This parameter can also be expressed as surface energy γ, where $G = 2\gamma$ for symmetric fracture.

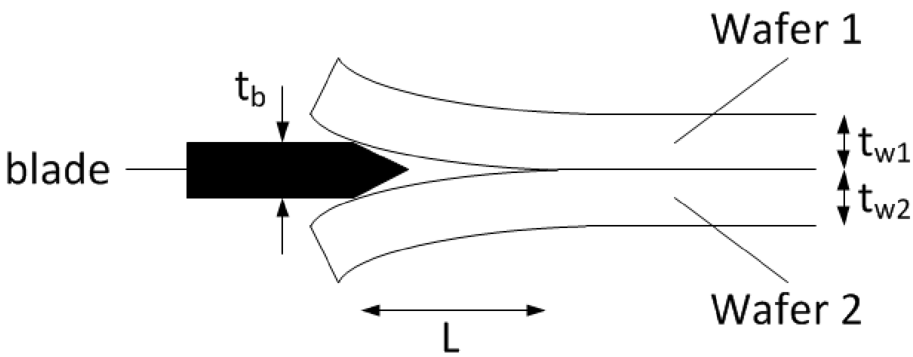

Figure 4.8: Illustration of the razor blade method. A razor blade is inserted between the bonded surfaces in order to quantify the bonding energy between them. From [88].

Theoretical Framework The energy release rate is calculated based on the energy balance between the elastic strain energy stored in the deformed substrates and the surface energy of the newly created surfaces during crack propagation. For bonded wafers with potentially different materials

Figure 4.9: Real example of the razor blade method. The de-bonded area can be identified by the presence of a Newton ring (light grey area). Adapted from [89].

and thicknesses, the energy release rate is given by [87]:

$$G = \frac{3t^2 E_1 E_2 h_1^3 h_2^3}{8L^2(E_1 h_1^3 + E_2 h_2^3)} \tag{4.2}$$

where t is the blade thickness, E_1 and E_2 are the Young's moduli of the substrates, h_1 and h_2 are the substrate thicknesses, and L is the crack length measured from the blade insertion point. For identical substrates ($E_1 = E_2 = E$ and $h_1 = h_2 = h$), this expression simplifies to:

$$G = \frac{3t^2 E h^3}{8L^2} = \frac{3t^2 E}{32y^2} \tag{4.3}$$

where $y = L/(2h)$ is the normalized crack length.

Experimental Procedure and Considerations The DCB measurement requires careful attention to several experimental parameters. First, crack length observation should be performed after a sufficient waiting period (typically 210 s) following blade insertion to avoid the effects of static fatigue at the bonding interface [87]. For transparent substrates, crack propagation can be observed optically; for opaque materials such as silicon, infrared imaging or acoustic microscopy is necessary.

Water Stress Corrosion Effects A critical consideration in bonding strength measurement, particularly for Si-based materials, is the effect of *water stress corrosion*. This phenomenon occurs when Si–O–Si bonds under applied stress react with water molecules according to the reaction [90]:

$$2\ \mathrm{SiO_2} + \mathrm{H_2O} \rightarrow 2\ \mathrm{Si\text{–}OH} + \frac{3}{2}\ \mathrm{O_2} \tag{4.4}$$

This water-assisted bond breakage significantly lowers the measured fracture toughness of SiO_2 structures, with reductions of up to 50% observed in wet atmospheres compared to dry conditions [90].

Controlled Atmosphere Measurements Recent advances have enabled the separation of water stress corrosion effects from ambient water and interfacial residual water through measurements under controlled atmospheres [87]. By performing DCB tests under three distinct conditions—wet atmosphere (ambient air with 50–70% relative humidity), dry atmosphere (pure N_2 gas), and vacuum ($\sim 3 \times 10^{-3}$ Pa)—the individual contributions can be isolated.

Figure 4.10 illustrates the water stress corrosion mechanism under different measurement atmospheres. In wet atmosphere, ambient water molecules attack the Si–O–Si bonds at the crack tip, reducing the measured bonding strength. In dry atmosphere, only residual interfacial water (trapped during low-temperature bonding) contributes to stress corrosion. Under vacuum conditions, interfacial water is partially desorbed as the crack propagates, minimizing the corrosion effect and yielding the highest measured bond strength [87].

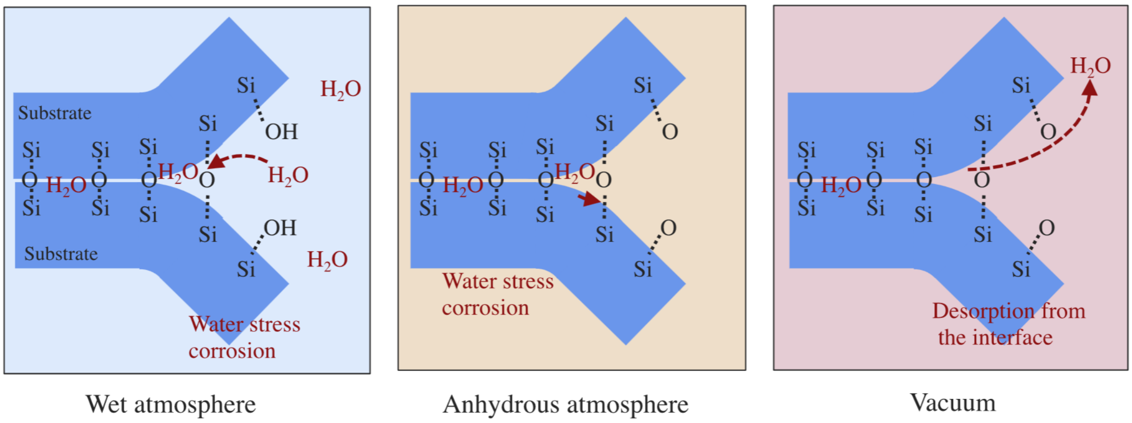

Figure 4.10: Schematic representation of water stress corrosion at the hydrophilic bonding interface during debonding under controlled atmospheres. Left: wet atmosphere showing ambient water attack on Si–O–Si bonds. Center: anhydrous (dry) atmosphere with reduced water availability. Right: vacuum conditions with partial desorption of interfacial water. Adapted from [87].

Advantages and Limitations The DCB method offers several important advantages: experimental simplicity requiring minimal equipment, direct measurement of surface energy, and applicability to full wafer-level bonding without the need for dicing into smaller specimens [87]. The technique is particularly valuable for process development and quality control in low-temperature bonding applications. However, several limitations must be considered:

- **Measurement range:** The method is limited to bond strengths lower than the intrinsic surface energy of the substrates, typically below 2–3 J m^{-2} for most Si-based materials [87].

- **Atmospheric sensitivity:** Results are strongly influenced by measurement atmosphere, with variations up to 50% between wet and dry conditions for SiO_2-based interfaces [90].
- **Edge effects and stress concentrations:** Crack initiation at wafer edges can introduce measurement artifacts.

4.5 Bonding Strength Test

One of the most widely used methods to assess bonding strength is the *tensile test* (also known as the pull test or direct tensile method). In this approach, bonded substrates are attached to two separate fixtures using a high-strength adhesive. These fixtures are then pulled apart in a direction perpendicular to the bonding interface, applying tensile stress that tends to separate the bonded wafers. A typical experimental setup is shown in Figure 4.11 [91, 92].

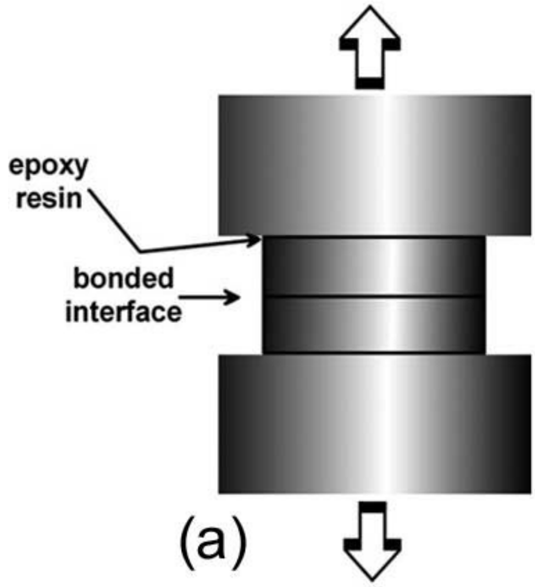

Figure 4.11: Schematic diagram of the tensile test setup for bonding strength measurement. The bonded wafer pair is glued to metal fixtures and subjected to perpendicular tensile force. Adapted from [23].

Fracture Mechanics Analysis Assuming that the bonding strength is lower than the adhesive's strength, cracks will be produced at the bonding interface when a certain strength threshold is exceeded. The fracture behavior can be characterized using linear elastic fracture mechanics (LEFM). Depending on the crack length, a *stress intensity factor* (K_I) can be estimated using the following expression [91, 23]:

$$K_I = \sigma Y \left(\frac{a}{\omega}\right) \sqrt{\pi a} \tag{4.5}$$

where a is the crack length, σ is the applied stress, and ω is the substrate width. The dimensionless parameter $Y\left(\frac{a}{\omega}\right)$ is a geometric correction factor that accounts for the finite geometry of the wafer and the crack position. For standard edge cracks (the most common failure mode), $Y \approx 1.12$ [93]. For internal cracks located far from the edge, $Y \approx 1.0$ [93].

Crack Propagation Criterion Cracks will propagate along the interface —and therefore cause complete fracture— if the stress intensity factor exceeds the material's fracture toughness limit ($K_I > K_{IC}$). This critical value often depends on several parameters, including the crystallographic orientation of the material, temperature, loading rate, and environmental conditions [91, 94].

For brittle materials under mode I loading (tensile opening), the fracture toughness can be theoretically estimated in terms of the surface energy (γ) as [23]:

$$K_{IC} = \sqrt{\frac{2E\gamma}{(1-\nu^2)}} \tag{4.6}$$

where E is the Young's modulus and ν is the Poisson ratio of the material.

From Equations 4.5 and 4.6, a critical stress (σ_C) for crack propagation can be derived:

$$\sigma_C = \sqrt{\frac{2E\gamma}{Y^2\left(\frac{a}{\omega}\right)\pi a(1-\nu^2)}} \tag{4.7}$$

This critical stress represents the bonding strength that can be experimentally measured.

Advantages and Limitations The tensile test offers several advantages, including experimental simplicity, direct measurement of bonding strength, and straightforward interpretation based on well-established fracture mechanics principles [92]. However, several limitations must be considered:

- **High measurement variance:** Real experiments exhibit significant scatter in bonding strength values, which can be attributed to local variations in surface parameters such as roughness, total thickness variation (TTV), particle contamination, and bonding defects [92, 23].
- **Edge effects:** Stress concentrations at wafer edges can lead to premature failure that does not accurately represent the average bonding quality [91].
- **Adhesive influence:** The test results may be affected by the properties of the adhesive used to attach the fixtures, particularly if the adhesive fails before the bond [92].
- **Sample preparation:** Careful alignment is crucial to ensure purely tensile loading; any misalignment introduces bending moments that affect the results.

Therefore, while the tensile test provides a useful estimate of bonding strength, results should be interpreted as indicative values rather than absolute measurements. Multiple samples and complementary characterization methods (such as the razor blade test) are recommended for comprehensive bonding quality assessment. **INCLUDE OUR OWN EXPERIMENT HERE**

4.6 Leak Tests using Spectroscopy

Once the cell is sealed and activated with the alkali metal vapor (typically Rb or Cs), leak tests can be performed over time to estimate its hermeticity. Leak tests involving laser absorption spectroscopy have been reported in literature [95, 96, 97, 98, 99]. In these tests, a tunable laser interacts with the alkali atoms. By measuring the width and central frequency of the resulting absorption profile, the internal environment of the cell can be monitored.

The optical transitions of alkali metals are characterized by the so-called D-line doublet. This fine structure arises from the spin-orbit interaction, which splits the first excited state (nP) into two distinct energy levels: $n^2P_{1/2}$ and $n^2P_{3/2}$. The transition from the ground state $n^2S_{1/2}$ to the lower excited state $n^2P_{1/2}$ is known as the D1 line, while the transition to the higher state $n^2P_{3/2}$ is the D2 line. These transitions are illustrated in Figure 4.12.

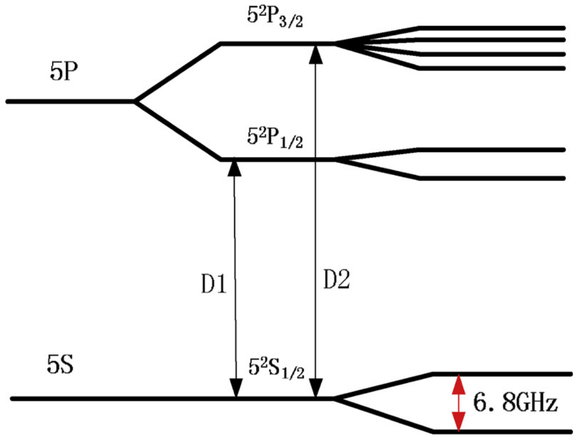

Figure 4.12: Energy level diagram showing the D1 and D2 transitions in an alkali metal atom. From [99]

Collisions between the alkali atoms and other gas molecules perturb the atomic energy levels, leading to collision-induced broadening and a frequency shift. If the cell's hermeticity is compromised and atmospheric gases leak into the cell, the resulting increase in internal pressure (P) will cause a detectable width broadening and a frequency shift of the Lorentzian profile. To quantify the gas pressure, the following relations are used:

$$\delta\nu = \delta P \tag{4.8}$$

$$\Delta\nu_L = \gamma P + \gamma_N \tag{4.9}$$

The first equation relates the frequency shift ($\delta\nu$) to the pressure (P) through the shift rate constant (δ). The second equation links the pressure to the Lorentzian full width at half maximum ($\Delta\nu_L$), involving the broadening rate constant γ and the natural linewidth γ_N.

Both δ and γ depend on the specific alkali-gas pair and the temperature, with values documented in literature [95, 96, 97, 98, 99]. Furthermore, the broadening rate γ at a temperature T_2 can be estimated from a known value at T_1 using:

$$\gamma(T_2) = \gamma(T_1)\left(\frac{T_1}{T_2}\right)^{\frac{1}{2}} \tag{4.10}$$

Note that the shift constant δ does not follow such a simple scaling law due to its more complex temperature dependence [95]

4.6.1 Experimental Proposal

The spectroscopic framework described above can be directly applied to assess the hermeticity of the fabricated vapor cells. Here, an experimental procedure similar to the one followed by Maurice et al. [100] is described.

First, the sealed and activated cell is heated to a nominal operating temperature T_{op} —chosen so that the alkali vapor pressure is sufficient to produce a measurable absorption signal— and held there throughout the measurement campaign using a temperature-stabilized mount. Temperature stability is critical, since any drift in T_{op} would introduce a systematic error in the extracted pressure through the temperature dependence of γ and δ [95].

A tunable laser is scanned across the D1 or D2 absorption line and the transmitted intensity is recorded with a photodetector. To avoid power broadening, the probe beam intensity is kept well below the saturation intensity of the transition. The resulting spectrum is fitted to a Voigt profile, from which the Lorentzian width $\Delta\nu_L$ and the line center ν_0 are extracted. Since the Doppler width depends only on T_{op} and the atomic mass, the Lorentzian component is unambiguously separated. The internal pressure is then retrieved using Equations 4.8 and 4.9, evaluated at T_{op} using literature values of γ and scaled if necessary via Equation 4.10. The two estimates of P —one from the width and one from the frequency shift— serve as a mutual consistency check.

Spectra are recorded at regular time intervals: initially daily, and subsequently weekly once short-term stability has been confirmed. The pressure $P(t)$ is plotted as a function of time and, in the case of a small but non-zero leak, a linear fit to $P(t)$ yields the leak rate $\dot{P}$. If no detectable increase is observed, the measurement noise propagated through Equation 4.9 sets an upper bound on the leak rate:

$$\sigma_P = \frac{\sigma_{\Delta\nu_L}}{\gamma(T_{\text{op}})} \tag{4.11}$$

where $\sigma_{\Delta\nu_L}$ is the uncertainty on the fitted Lorentzian width. This bound provides a physically meaningful hermeticity statement even in the absence of a detectable leak, and allows direct comparison with benchmarks from the literature: the 707-day vacuum integrity demonstrated by Artusio-Glimpse et al. [33] using Rydberg EIT spectroscopy, and the short-term seal integrity inferred from CPT resonance stability by Kobtsev et al. [101] over timescales up to $\sim$1000 s.

Chapter 5

Experiments & Applications

5.1 Corrosion Inspection for Railway Tracks

A new generation of highly sensitive and precise magnetic sensors based on vapor cells has recently transitioned from development to the commercial stage. These devices, known as *Optically Pumped Magnetometers* (OPMs), can detect extraordinarily weak magnetic fields, with sensitivities reaching the pico-tesla ($\mathrm{pT}/\sqrt{\mathrm{Hz}}$) or even femto-tesla ($\mathrm{fT}/\sqrt{\mathrm{Hz}}$) scale. Crucially for practical applications, they maintain this extreme precision while operating within a wide dynamic range, allowing them to function reliably even in unshielded environments [102, 103].

The core operating principle of these sensors relies on using laser light to manipulate and measure the quantum spin states of an alkali metal gas (typically ^{87}Rb or ^{133}Cs) enclosed in a glass cell [104]. This physical process occurs in three consecutive stages, beginning with the preparation of the atomic state through optical pumping. During this first stage, a circularly polarized (σ^+) laser beam, called the pump beam, passes through the vapor and transfers its angular momentum to the alkali atoms, thereby aligning their spins in a specific direction. Concurrently, a constant magnetic field (B_{DC}) is applied parallel to this laser to guide and stabilize the aligned atoms into a baseline energy state or Zeeman sublevel.

Once the atoms are prepared, the second stage involves their resonant excitation by means of an alternating radio-frequency magnetic field (B_{RF}) applied perpendicularly to the baseline field. When the frequency of this radio-frequency field matches the natural precession frequency of the atoms—known as the Larmor frequency ($\omega_L = \gamma B_{\mathrm{tot}}$, where γ is the gyromagnetic ratio)—it drives a coherent transition in the atomic spins, forcing them to rotate synchronously in circles.

Finally, the third stage consists of the optical interrogation and electronic demodulation of the system. To detect the atomic rotation without destroying the prepared quantum state, a second, linearly polarized laser beam, designated as the probe beam, is sent through the cell. As this probe light passes through the spinning atoms, its plane of polarization undergoes a periodic rotation due to the paramagnetic Faraday effect, oscillating at the exact frequency of the Larmor precession. A balanced polarimeter captures this oscillating light and converts it into an alternating electrical signal (AC). This signal is subsequently processed by a lock-in amplifier, which demodulates the AC signal using the radio-frequency field as a reference, yielding precise direct current (DC) outputs for both amplitude and phase. With this configuration, any shifts in the local magnetic field disrupt the atomic resonance, producing an immediate, proportional change in these DC voltage outputs.

The schematic layout of this radio-frequency OPM configuration, including the orthogonal arrangement of the laser beams and the magnetic fields, is illustrated in Figure 5.1.

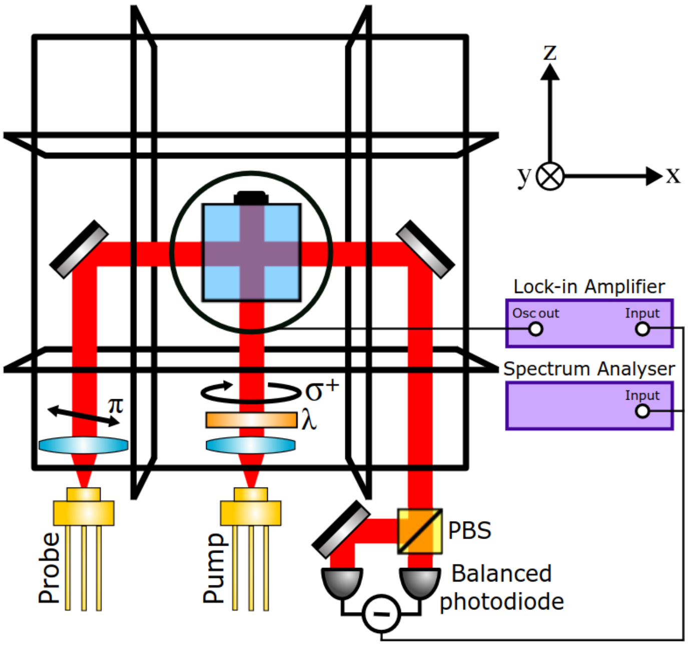

Figure 5.1: Schematic layout of a radio-frequency OPM. The alkali vapor is spin-polarized along the static field (B_{DC}) by a σ^+ pump beam. An orthogonal AC magnetic field excites the Larmor precession, which rotates the polarization plane of a perpendicular, linearly polarized probe beam. This optical rotation is detected by a polarimeter and analyzed using a lock-in amplifier. Adapted from [103].

Several non-destructive inspection (NDI) experiments have successfully demonstrated the use of atomic magnetometers for Electromagnetic Induction Imaging (EMI) [105] and Magnetic Induction Tomography (MIT) [106]. Recent studies [107, 108] have characterized the penetration capabilities of these devices through structural barriers. It has been established that OPMs can achieve high-fidelity object imaging through thick conductive enclosures under unshielded, room-temperature conditions without requiring background subtraction, as shown in Figure 5.2. Moreover, fully portable scanning heads equipped with active-compensation coils have been demonstrated, proving capable of operating within complex, real-world magnetic environments [103].

When microfabricated into advanced architectures such as LWVCs, OPMs can achieve ultra-compact footprints at the MEMS scale with minimal power requirements. By eliminating the bulky cryogenic cooling systems inherent to conventional Superconducting Quantum Interference Devices (SQUIDs), these miniaturized atomic sensors enable highly portable operations and scalable, multi-channel array configurations.

5.1.1 Application Scenario: Railway Track Corrosion

UNFINISHED. CHECK VALUES. CHECK MESHING. A novel application of these devices is presented in this section. The objective is to utilize an OPM integrated with an LWVC sensor

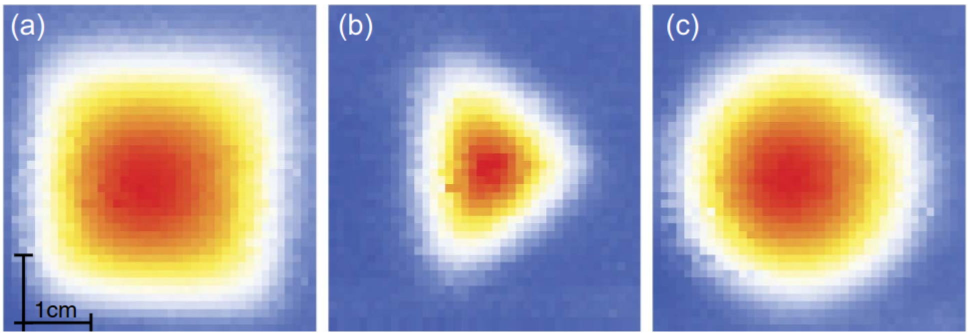

Figure 5.2: OPM imaging of (a) a $37\,\mathrm{mm}\times37\,\mathrm{mm}$ **square, (b) an isosceles triangle with one side of** $37\,\mathrm{mm}$ **and two sides of** $30\,\mathrm{mm}$**, and (c) a disk with a** $37\,\mathrm{mm}$ **diameter. Adapted from [106].**

head to detect the onset and progression of structural corrosion at the foot of railway tracks.

The present work employs a QuSpin Total-Field Magnetometer as the reference OPM sensor. This is a pulsed laser-pumped rubidium free-induction-decay device operating on the principles described above. Its key performance specifications are summarized in Table 5.1. Of particular relevance to the present application is the scalar sensitivity of $< 3\,\mathrm{pT}/\sqrt{\mathrm{Hz}}$, the $500\,\mathrm{Hz}$ measurement bandwidth at $1000\,\mathrm{samples/s}$, and the dynamic range of 1–$150\,\mu\mathrm{T}$, which covers the total field environment encountered in unshielded railway settings. The sensor head weighs only $12\,\mathrm{g}$ and occupies $17.7 \times 19.8 \times 35.8\,\mathrm{mm}^3$, making it compatible with compact, portable scanning assemblies [102].

Table 5.1: Key technical specifications of the QuSpin Zero-Field Magnetometer used as the reference OPM sensor in this work [102].

Parameter	**Value**
Sensor type	Pulsed laser-pumped Rb, free induction decay
Scalar sensitivity	$< 3\,\mathrm{pT}/\sqrt{\mathrm{Hz}}$
Vector sensitivity (optional)	$< 0.1\,\mathrm{nT}/\sqrt{\mathrm{Hz}}$
Bandwidth	$500\,\mathrm{Hz}$
Data rate	$1000\,\mathrm{samples/s}$
Dynamic range	1000–$150{,}000\,\mathrm{nT}$
Deadzone	Axial only, $< \pm 7^\circ$ cone about Earth's field
Heading error	$< 3\,\mathrm{nT}$ (uncompensated)
Max. gradient field	$300\,\mathrm{nT/cm}$
Operating temperature	-15 to $+55\,^\circ\mathrm{C}$
Sensor head size	$17.7 \times 19.8 \times 35.8\,\mathrm{mm}$
Sensor head mass	$12\,\mathrm{g}$
Power consumption	$2.5\,\mathrm{W}$ ($3.5\,\mathrm{W}$ startup)
Output interface	UART (RS-232), USB

A standard railway track profile (Figure 5.3) is analyzed. These tracks are manufactured from high-strength pearlitic carbon steel, with high electrical conductivity ($\sigma_{\mathrm{rail}} \approx 4.0$–$5.0 \times 10^6\,\mathrm{S/m}$) and strong ferromagnetic permeability ($\mu_{r,\mathrm{rail}} \approx 50$–$200$). When exposed to atmospheric oxygen and moisture, electrochemical oxidation produces a degradation layer of iron oxides and hydroxides (primarily Fe_2O_3, Fe_3O_4, and FeOOH) at the rail foot. This corrosion layer presents a dramatically lower electrical conductivity ($\sigma_{\text{corr}} \approx 10^{-2}$–$10^{2}$ S/m) and a relative permeability that drops to near unity ($\mu_{r,\text{corr}} \approx 1$–$5$), rendering it effectively paramagnetic.

Figure 5.3: Cross-sectional schematic of a standard railway track, emphasizing the rail foot region susceptible to atmospheric corrosion.

The detection principle is as follows. An excitation coil driven by an alternating current generates a time-varying primary magnetic field $B_1(t) = B_1 e^{i\omega t}$, which penetrates the rail and induces eddy currents ($\mathbf{J} = \sigma\mathbf{E}$) in the bulk steel according to Faraday's law. The amplitude of these currents decays exponentially with depth, characterized by the skin depth:

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma}} = \sqrt{\frac{1}{\pi f \mu_0 \mu_r \sigma}}, \tag{5.1}$$

where f is the excitation frequency, μ_0 is the vacuum permeability, μ_r is the relative permeability, and σ is the electrical conductivity of the domain. Because the corrosion layer presents a severe contrast in both μ_r and σ relative to healthy steel, the eddy current distribution is perturbed in the vicinity of the corroded region, altering the secondary field B_2 that it generates. The total field $B_{\text{tot}} = B_1 + B_2$ thus carries a signature of the corrosion that can be resolved by the OPM.

5.1.2 Finite-Element Simulation

To evaluate the feasibility of this concept prior to experimental implementation, a comprehensive 2D finite-element method (FEM) simulation was carried out in COMSOL Multiphysics, using the *AC/DC Module* (*Magnetic Fields* physics interface) in the frequency-domain formulation. Two scenarios are modelled: a healthy rail (benchmark) and a defective rail containing a localized corrosion layer of variable depth at the foot (Figure 5.4).

Geometry and boundary conditions. The simulation domain consists of the 2D cross-section of a standard rail profile, surrounded by a sufficiently large air region ($\sim$10 rail widths on each side) to avoid artefacts from the outer boundary. The outer boundary of the air domain is assigned a magnetic insulation condition ($\mathbf{n} \times \mathbf{A} = 0$, where $\mathbf{A}$ is the magnetic vector potential), which is appropriate for a magnetically open domain at distances far from the source. The excitation coil is modelled as a single rectangular conductor carrying a prescribed current density $\mathbf{J}_e$ (ampere-turns per unit area).

Mesh strategy. Accurate resolution of the electromagnetic fields requires the mesh to respect the skin depth constraint at each frequency. In the steel and corrosion domains, the maximum element size is set to $\delta/3$ to ensure at least three elements per skin depth, following standard FEM practice for eddy-current problems [**comsol_ref**]. Since δ varies from $\sim$7 mm at $10\,\text{Hz}$ to $\sim$0.07 mm at $100\,\text{kHz}$ for the rail steel, frequency-dependent adaptive meshes are generated for each simulation point in the parameter sweep. In the air region, where no skin effect applies, a free triangular mesh with maximum element size constrained to one tenth of the smallest geometric feature is used. The corrosion layer is meshed with a boundary layer refinement of five inflation layers at its interfaces with both the steel and the air gap. The resulting meshes contain between 8×10^4 and 3×10^5 elements depending on frequency, with the highest element counts at high frequencies where δ is smallest.

Material assignments and parametric sweep. The rail steel is assigned $\sigma_{\text{rail}} = 4.5 \times 10^6\,\text{S/m}$ and $\mu_{r,\text{rail}} = 100$ (representative values for pearlitic rail steel). The corrosion layer is assigned $\sigma_{\text{corr}} = 1\,\text{S/m}$ and $\mu_{r,\text{corr}} = 2$. The simulation sweeps excitation frequencies from $10\,\text{Hz}$ to $100\,\text{kHz}$ over a logarithmic grid of 30 points per decade, and corrosion layer depths from 0 (healthy) to $5\,\text{mm}$ in $0.5\,\text{mm}$ steps, under three representative coil current strengths.

Figure 5.4: Geometric configuration of the 2D COMSOL simulation domain, indicating the excitation coil position, the corrosion zone at the rail foot, and the mandatory clearance line above which the OPM sensor must operate.

Evaluation cutpoints and operational constraints. The simulation imposes two simultaneous operational constraints on any candidate sensor position:

1. **Dynamic range boundary:** The total field B_{tot} must not exceed the QuSpin linear operating range of $150{,}000\,\text{nT}$ to prevent sensor saturation.

2. **Sensitivity boundary:** The differential field between healthy and corroded states must satisfy $\Delta B = |B_{\text{healthy}} - B_{\text{corroded}}| \geq 3\,\text{pT}$, the scalar noise floor of the sensor.

Three cutpoints above the mechanical clearance line are evaluated: $P_1(x_1, y_1)$, $P_2(x_2, y_2)$, and $P_3(x_3, y_3)$, at increasing heights above the rail surface.

5.1.3 Results

Figure 5.5 shows the computed total field maps B_{tot} for both the healthy and corroded scenarios at the optimal operating frequency. Figure 5.6 presents the pointwise difference $\Delta B_{\text{2D}} = |B_{\text{healthy}} - B_{\text{corroded}}|$ over the full simulation plane, making the spatial extent and magnitude of the corrosion signature immediately visible.

(a) Healthy rail. **(b) Corroded rail foot.**

Figure 5.5: 2D COMSOL maps of the total magnetic field B_{tot} for both structural conditions at the optimal excitation frequency.

Figure 5.6: Pointwise differential field map $\Delta B_{\text{2D}} = |B_{\text{healthy}} - B_{\text{corroded}}|$ over the full simulation domain at the optimal frequency. The spatial extent of the corrosion signature and the field enhancement in the vicinity of the defect are clearly visible.

Figure 5.7 shows ΔB extracted at the three cutpoints as a function of frequency for the maximum corrosion depth considered.

Figure 5.7: Differential magnetic field amplitude $\Delta B = |B_{\text{healthy}} - B_{\text{corroded}}|$ at cutpoints P_1, P_2, and P_3 as a function of excitation frequency. The horizontal dashed line marks the QuSpin noise floor of $3\,\text{pT}$.

The results reveal a characteristic trade-off governed by the frequency-dependent skin depth. At low frequencies ($f < 100\,\text{Hz}$), the field penetrates the full depth of the rail foot, but the slow rate of flux change produces weak eddy currents and a differential response ΔB near or below the sensor noise floor. At high frequencies ($f > 50\,\text{kHz}$), the skin effect confines the field to a thin surface layer ($\delta \to 0$), preventing interaction with the interior of the corrosion zone.

An optimal operating window is identified in the intermediate band of 1–20 kHz, where the skin depth matches the spatial scale of the expected corrosion layers. Within this band, ΔB peaks sharply and, at cutpoints P_1 and P_2, successfully exceeds the 3 pT threshold while the background field remains below the $150{,}000\,\text{nT}$ dynamic range limit of the QuSpin sensor. These findings validate the technical feasibility of using unshielded, compact LWVC-based OPMs for non-contact, real-time railway track corrosion monitoring, providing a clear path toward automated structural health evaluation systems.

5.2 Magnetoencephalography

5.3 Magnetometry Experiments on Vapor Cells

CAMBIAR NOM A ZERO FIELD RESONANCE O ALGO AIXINA MIRAR PAPER DE GIANVITO, AL FINAL SE EXPLICA BE LA ZERO FIELD RESONANCE In this experiment, performed at the facilities of the Physics Department at the *Politecnico di Bari* (POLIBA) in Bari, Italy, the goal was to demonstrate the capacity of vapor cells to serve as precise magnetometers in a laboratory setting. As described in [14, 104], when circularly polarized light interacts with alkali atoms (such as Rubidium or Cesium), it *transfers* some of its angular momentum into the gas, effectively generating a non-zero electron spin polarization state inside the vapor cell.

In the presence of an external magnetic field $\vec{B}$, the atomic spins undergo Larmor precession at the frequency:

$$\omega_L = \gamma B \tag{5.2}$$

where γ is the gyromagnetic ratio ($2\pi \times 3.5\,\text{kHz/G}$ for ^{133}Cs). This precession modulates the projection of the atomic spin along the probe beam direction, causing periodic variations in the optical absorption and thus in the transmitted light intensity detected by the photodiode. When a linear magnetic field ramp—such as $B(t) = B_0 + \alpha t$—is applied across the vapor cell, the transmitted light intensity exhibits a Lorentzian resonance peak centered at $B = 0$. This phenomenon, known as the *ground-state Hanle resonance* or *zero-field resonance*, arises from the following mechanism.

For $B \neq 0$, the total magnetic field causes the spin polarization to precess around $\vec{B}$ at the Larmor frequency $\omega_L = \gamma B$. This precession continuously rotates the spin orientation away from the pump beam axis, reducing the time-averaged spin projection along the beam and thus increasing absorption. At $B = 0$ there is no precession; the spin polarization is maintained along the pump axis, minimizing absorption and producing a transmission maximum.

By analyzing the Lorentzian profile, properties of the vapor cell such as the spin coherence lifetime τ can be determined. The relationship between the Lorentzian full width at half maximum (FWHM) and τ is given by:

$$\Delta B_{\text{FWHM}} \sim \frac{1}{\gamma \tau} \tag{5.3}$$

5.3.1 Experimental Setup

A Twinleaf vapor cell filled with cesium and nitrogen as a buffer gas was placed inside a magnetic shield to isolate it from background fields, including Earth's magnetic field. A laser tuned to the D1 transition of Cs ($\lambda = 895$ nm) was directed into the cell. A half-wave plate (HWP) and a polarizing beam splitter (PBS) were used to control the optical power, and a quarter-wave plate (QWP) placed just before the magnetic shield converted the beam to circular polarization.

Inside the magnetic shield, coils generate an alternating magnetic field ramp ($B_x = B_0 + \alpha t$) along the beam axis (X-axis). This time-dependent field produces a time-dependent absorption profile, detected by a photodetector placed outside the shield and connected to a lock-in amplifier. Additional DC fields were applied along the Y- and Z-axes to compensate residual transverse fields. The experiment was carried out at room temperature. The setup is illustrated in Figure 5.8.

5.3.2 Results and Analysis

Measurements were taken under different DC field configurations along the Y- and Z-axes to study their effects on the Lorentzian peak amplitude, width, and noise level. Figure 5.9 compares the fitted Lorentzian profiles obtained under these different configurations.

From these results it is clear that the DC field configuration significantly affects the sharpness of the observed resonance. Signal amplitudes below 10 mV are obtained for the least optimal configuration ($B_y = 3600$ nT, $B_z = 0$ nT), while amplitudes greater than 40 mV are achieved with the configuration ($B_y = 0$ nT, $B_z = 4815$ nT). For this specific setup, applying a DC field along the Y-axis reduces the resonance amplitude and increases the FWHM, whereas a DC field along the Z-axis has the opposite effect.

The optimal configuration ($B_y = 0$ nT, $B_z = 4815$ nT) was analyzed further. Figure 5.10 shows the raw data and Lorentzian fit for this configuration. The resonance amplitude is 42.2 mV and the estimated FWHM is 8726.3 nT. The Lorentzian maximum is displaced from the origin to $B_x = 865.5$ nT; this systematic offset is attributable to a residual magnetic field along the X-axis

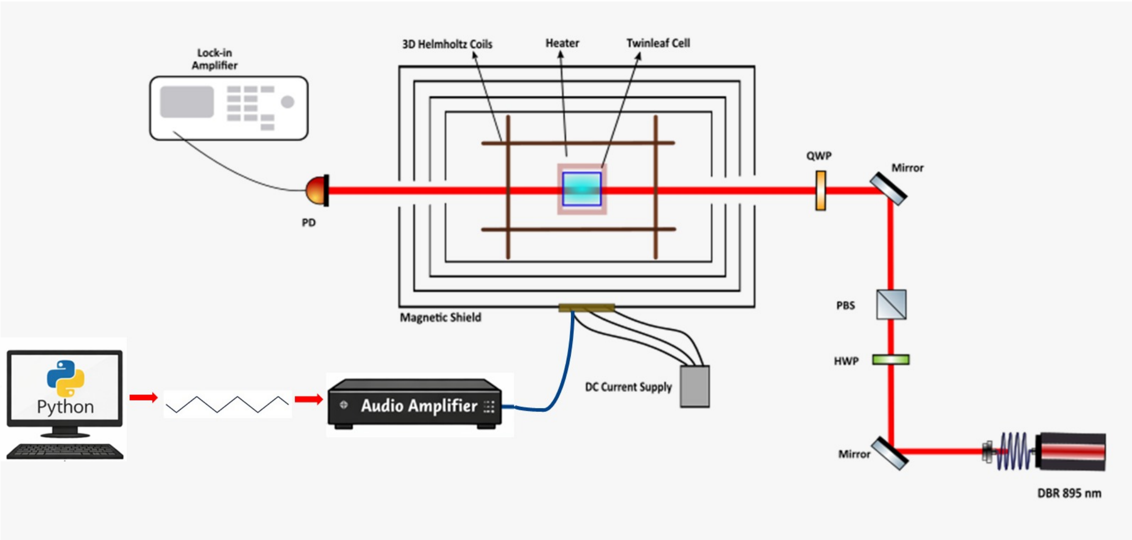

Figure 5.8: Experimental setup used to detect the ground-state Hanle resonance in a cesium vapor cell.

that was not fully compensated by the shielding or the transverse DC coils, effectively shifting the zero-field condition away from $B_x = 0$.

Applying Eq. (5.3) with $\gamma = 2\pi \times 3.5\,\mathrm{kHz/G} = 2\pi \times 0.035\,\mathrm{Hz/nT}$ and $\Delta B_{\mathrm{FWHM}} = 8726.3\,\mathrm{nT}$ yields a spin coherence lifetime of:

$$\tau = \frac{1}{\gamma\,\Delta B_{\mathrm{FWHM}}} = \frac{1}{2\pi \times 0.035 \times 8726.3} \approx 0.52\,\mathrm{ms} \tag{5.4}$$

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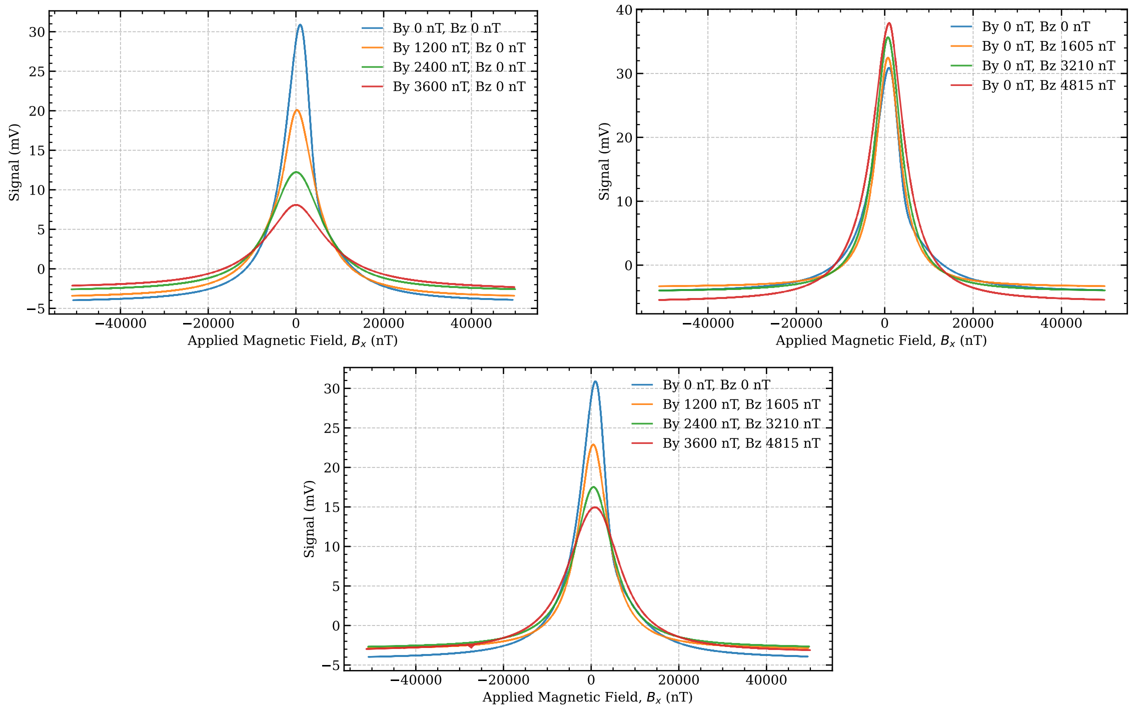

Figure 5.9: Lorentzian profiles for different DC field configurations: (a) B_y varied with $B_z = 0$, (b) B_z varied with $B_y = 0$, and (c) B_y and B_z varied simultaneously.

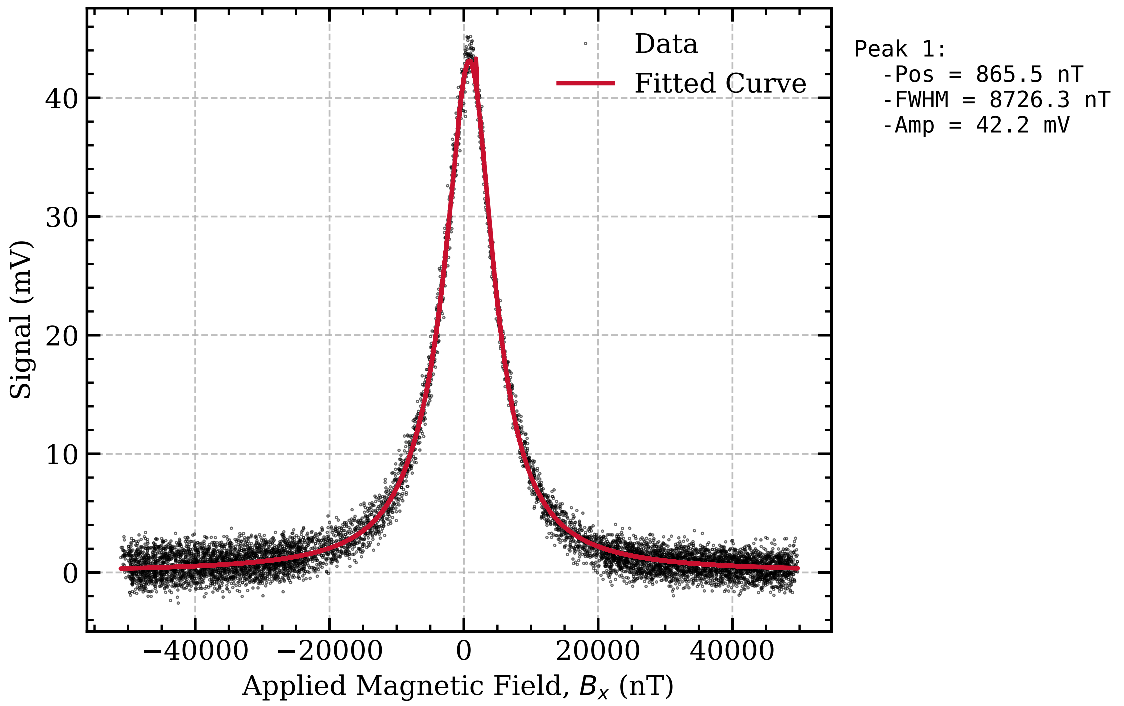

Figure 5.10: Raw data and Lorentzian fit of the ground-state Hanle resonance for the optimal DC field configuration ($B_y = 0\,\mathrm{nT}$, $B_z = 4815\,\mathrm{nT}$). The resonance amplitude is $42.2\,\mathrm{mV}$, the FWHM is $8726.3\,\mathrm{nT}$, and the derived spin coherence lifetime is $\tau \approx 0.52\,\mathrm{ms}$.

Chapter 6

Conclusions

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Part II

Annex

Appendix A

Recipes in Literature

A.1 Recipes in literature

Table A.1: Summary of chemical cleaning procedures, surface activation methods, and resulting bonding qualities.

Reference	Chemical Cleaning Steps							Surface Activation	Post-Treatment	Bonding Strength
	1	2	3	4	5	6	7			
Domenico Tuli Thesis	DIW 10’ 25°C	—	—	—	—	—	—	—	—	0.06 J/m^2
	Acetone 10’ 25°C	—	—	—	—	—	—	Plasma 30s 15W 35mTorr	—	0.06 J/m^2
	Acetone 10’ 25°C	DIW 10’ 25°C	Micro Soap 10’ 25°C	DIW 10’ 25°C	—	—	—	—	—	0.06 J/m^2
	Acetone 10’ 25°C	DIW 10’ 25°C	RCA1 10’ 60°C	DIW 10’ 25°C	—	—	—	—	—	0.06 J/m^2
	Acetone 10’ 25°C	DIW 10’ 25°C	RCA1 10’ 60°C	NH_4OH 2’ 25°C	—	—	—	—	—	0.06 J/m^2
	Acetone 10’ 25°C	DIW 10’ 25°C	Micro Soap 10’ 25°C	RCA1 10’ 60°C	DIW 10’ 25°C	—	—	—	—	0.06 J/m^2
	Acetone 10’ 25°C	DIW 10’ 25°C	Micro Soap 10’ 25°C	RCA1 10’ 60°C	DIW 10’ 25°C	—	—	—	—	0.1 J/m^2
	Acetone 10’ 25°C	DIW 10’ 25°C	Micro Soap 10’ 25°C	RCA1 10’ 60°C	NH_4OH 2’ 25°C	—	—	—	—	0.1 J/m^2
	Acetone 10’ 25°C	DIW 10’ 25°C	Piranha 10’ 60°C	DIW 10’ 25°C	—	—	—	Plasma 30s 15W 35mTorr	—	0.1 J/m^2
	Acetone 10’ 25°C	DIW 10’ 25°C	Piranha 10’ 60°C	DIW 10’ 25°C	Micro Soap 10’ 25°C	RCA1 10’ 60°C	DIW 10’ 25°C	—	—	0.1 J/m^2
	Acetone 10’ 25°C	DIW 10’ 25°C	Piranha 10’ 60°C	DIW 10’ 25°C	Micro Soap 10’ 25°C	RCA1 10’ 60°C	NH_4OH 2’ 25°C	—	—	0.1 J/m^2
Cho et al. 2001	HF (dipped)	Piranha 15’	RCA1 15’	—	—	—	—	O_2 Plasma 15-120s 50-300W	Annealing 50h >105°C	2.5 J/m^2
Eichler et al. 2008	RCA1	RCA2	—	—	—	—	—	O_2 Plasma 40s 30W 1atm	Annealing 6h 120°C	2.2 J/m^2
Li et al. 2013	SC1 10’ 80°C	SC2 10’ 80s	DIW (rinse)	RCA1 10’ 60°C	—	—	—	O_2 Plasma 60s 50W 20mTorr	Press. 3x10^5 Pa → Annealing 6h 200-800°C	11 MPa BS
Wang et al. 2020	Acetone	Ethanol	DIW	—	—	—	—	O_2 Plasma 90s 200W 375mTorr	Wait 12h Room T → Annealing 12h 150°C	5.5 MPa BS
Takei et al. 2010	RCA1 10’ 70°C	DIW (rinse)	HF 2%	—	—	—	—	O_2 Plasma 30s 400W 900mTorr	Annealing 8h 250°C 5MPa	1 MPa BS
Lee et al. 2022	—	—	—	—	—	—	—	N_2 Plasma 15s 100W 100mTorr	DIW → Annealing 1-4h 200-500°C	2.5x w/o plasma
Suni et al. 2002	—	—	—	—	—	—	—	N_2 Plasma 30s 50-150W	RCA1 → DIW 5’ → Annealing → Dicing → Annealing	>2.5 J/m^2
	—	—	—	—	—	—	—	O_2 Plasma 45s 50-150W	Annealing 2h 100°C → Dicing → Annealing 2h 200°C	∼2.5 J/m^2
Park et al. 2025	Piranha	HF dip	—	—	—	—	—	O_2 Plasma 30s 800W 800mTorr	Press. 5’ 5000mBar → Annealing 1h 400°C	∼1.7 J/m^2
Our proc. (P11-P13)	Acetone 10’ 25°C	DIW 5’ + dry	Piranha 10’ 75°C	DIW 5’ + dry	RCA1 10’ 75°C	DIW 5’ + dry	—	O_2 Plasma 60s 30W 1200mTorr O_2 Plasma 120s 30W 1200mTorr O_2 Plasma 180s 30W 1200mTorr	DIW 5’ → Annealing >12h 150°C	TBD
Our proc. (P21-P23)	Acetone 10’ 25°C	DIW 5’ + dry	Piranha 10’ 75°C	DIW 5’ + dry	—	—	—	O_2 Plasma 60s 30W 1200mTorr O_2 Plasma 120s 30W 1200mTorr O_2 Plasma 180s 30W 1200mTorr	DIW 5’ → Annealing >12h 150°C	TBD